

Marriage and life satisfaction: T-Test

Notes

Output Created		06-NOV-2024 15:57:53
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	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.
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Group Statistics

	married & remarried	N	Mean	Std. Deviation	Std. Error Mean
Total life satisfaction	1.00	218	23.57	6.676	.452
	2.00	218	21.19	6.667	.452

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means	
		F	Sig.	t	df
Total life satisfaction	Equal variances assumed	.039	.844	3.726	434
	Equal variances not assumed			3.726	433.999

Independent Samples Test

		t-test for Equality of Means		
		Significance		Mean Difference
		One-Sided p	Two-Sided p	
Total life satisfaction	Equal variances assumed	<.001	<.001	2.381
	Equal variances not assumed	<.001	<.001	2.381

Independent Samples Test

		t-test for Equality of Means		
		Std. Error Difference	95% Confidence Interval of the Difference	
			Lower	Upper
Total life satisfaction	Equal variances assumed	.639	1.125	3.637
	Equal variances not assumed	.639	1.125	3.637

Independent Samples Effect Sizes

				95% Confidence Interval	
Standardizer ^a			Point Estimate	Lower	Upper
Total life satisfaction	Cohen's d	6.671	.357	.167	.546
	Hedges' correction	6.683	.356	.167	.545
	Glass's delta	6.667	.357	.166	.547

a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Question2: Marriage and optimism levels: One-way Anova

Notes

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Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
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Resources	Processor Time	00:00:00.11
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Descriptives

Total Optimism

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound
SINGLE	104	21.51	4.520	.443	20.63
STEADY RELATIONSHIP	37	21.59	4.524	.744	20.09
LIVING WITH PARTNER	36	21.64	4.549	.758	20.10
MARRIED FIRST TIME	188	22.51	4.199	.306	21.90
REMARRIED	29	23.00	4.488	.833	21.29
SEPARATED	10	18.50	4.994	1.579	14.93
DIVORCED	24	22.67	4.280	.874	20.86
WIDOWED	7	25.57	4.315	1.631	21.58
Total	435	22.12	4.429	.212	21.70

Descriptives

Total Optimism

	95% Confidence Interval for Mean Upper Bound	Minimum	Maximum
SINGLE	22.39	7	30
STEADY RELATIONSHIP	23.10	10	29
LIVING WITH PARTNER	23.18	8	29
MARRIED FIRST TIME	23.11	9	30
REMARRIED	24.71	15	29
SEPARATED	22.07	10	28
DIVORCED	24.47	15	30
WIDOWED	29.56	16	29
Total	22.53	7	30

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Total Optimism	Based on Mean	.264	7	427	.967
	Based on Median	.442	7	427	.876
	Based on Median and with adjusted df	.442	7	405.166	.876
	Based on trimmed mean	.334	7	427	.938

ANOVA

Total Optimism

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	329.264	7	47.038	2.454	.018
Within Groups	8183.757	427	19.166		
Total	8513.021	434			

ANOVA Effect Sizes^{a,b}

		Point Estimate	95% Confidence Interval	
			Lower	Upper
Total Optimism	Eta-squared	.039	.001	.065
	Epsilon-squared	.023	-.015	.050
	Omega-squared Fixed-effect	.023	-.015	.049
	Omega-squared Random-effect	.003	-.002	.007

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Robust Tests of Equality of Means

Total Optimism

	Statistic ^a	df1	df2	Sig.
Welch	2.069	7	53.332	.063
Brown-Forsythe	2.318	7	118.374	.030

a. Asymptotically F distributed.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	Mean Difference (I-J)	Std. Error	Sig.	95% ... Lower Bound
SINGLE	STEADY RELATIONSHIP	-.085	.838	1.000	-2.64
	LIVING WITH PARTNER	-.129	.847	1.000	-2.71
	MARRIED FIRST TIME	-.996	.535	.579	-2.63
	REMARRIED	-1.490	.919	.737	-4.29
	SEPARATED	3.010	1.449	.432	-1.41
	DIVORCED	-1.157	.991	.941	-4.18
	WIDOWED	-4.062	1.709	.256	-9.27
STEADY RELATIONSHIP	SINGLE	.085	.838	1.000	-2.47
	LIVING WITH PARTNER	-.044	1.025	1.000	-3.17
	MARRIED FIRST TIME	-.911	.787	.943	-3.31
	REMARRIED	-1.405	1.086	.901	-4.71
	SEPARATED	3.095	1.560	.494	-1.66
	DIVORCED	-1.072	1.147	.983	-4.57
	WIDOWED	-3.977	1.804	.351	-9.47
LIVING WITH PARTNER	SINGLE	.129	.847	1.000	-2.45
	STEADY RELATIONSHIP	.044	1.025	1.000	-3.08
	MARRIED FIRST TIME	-.866	.796	.959	-3.29
	REMARRIED	-1.361	1.092	.918	-4.69
	SEPARATED	3.139	1.565	.479	-1.63
	DIVORCED	-1.028	1.154	.987	-4.54
	WIDOWED	-3.933	1.808	.369	-9.44
MARRIED FIRST TIME	SINGLE	.996	.535	.579	-.63
	STEADY RELATIONSHIP	.911	.787	.943	-1.49
	LIVING WITH PARTNER	.866	.796	.959	-1.56
	REMARRIED	-.495	.873	.999	-3.16
	SEPARATED	4.005	1.421	.093	-.32
	DIVORCED	-.161	.949	1.000	-3.05
	WIDOWED	-3.066	1.685	.607	-8.20
REMARRIED	SINGLE	1.490	.919	.737	-1.31
	STEADY RELATIONSHIP	1.405	1.086	.901	-1.90
	LIVING WITH PARTNER	1.361	1.092	.918	-1.97
	MARRIED FIRST TIME	.495	.873	.999	-2.17
	SEPARATED	4.500	1.605	.097	-.39
	DIVORCED	.333	1.208	1.000	-3.35
	WIDOWED	-2.571	1.844	.859	-8.19
SEPARATED	SINGLE	-3.010	1.449	.432	-7.42
	STEADY RELATIONSHIP	-3.095	1.560	.494	-7.85

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	95% ... Upper Bound
SINGLE	STEADY RELATIONSHIP	2.47
	LIVING WITH PARTNER	2.45
	MARRIED FIRST TIME	.63
	REMARRIED	1.31
	SEPARATED	7.42
	DIVORCED	1.86
	WIDOWED	1.15
STEADY RELATIONSHIP	SINGLE	2.64
	LIVING WITH PARTNER	3.08
	MARRIED FIRST TIME	1.49
	REMARRIED	1.90
	SEPARATED	7.85
	DIVORCED	2.42
	WIDOWED	1.52
LIVING WITH PARTNER	SINGLE	2.71
	STEADY RELATIONSHIP	3.17
	MARRIED FIRST TIME	1.56
	REMARRIED	1.97
	SEPARATED	7.91
	DIVORCED	2.49
	WIDOWED	1.58
MARRIED FIRST TIME	SINGLE	2.63
	STEADY RELATIONSHIP	3.31
	LIVING WITH PARTNER	3.29
	REMARRIED	2.17
	SEPARATED	8.33
	DIVORCED	2.73
	WIDOWED	2.07
REMARRIED	SINGLE	4.29
	STEADY RELATIONSHIP	4.71
	LIVING WITH PARTNER	4.69
	MARRIED FIRST TIME	3.16
	SEPARATED	9.39
	DIVORCED	4.01
	WIDOWED	3.04
SEPARATED	SINGLE	1.41
	STEADY RELATIONSHIP	1.66

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	Mean Difference (I-J)	Std. Error	Sig.	95% ... Lower Bound
	LIVING WITH PARTNER	-3.139	1.565	.479	-7.91
	MARRIED FIRST TIME	-4.005	1.421	.093	-8.33
	REMARRIED	-4.500	1.605	.097	-9.39
	DIVORCED	-4.167	1.648	.186	-9.19
	WIDOWED	-7.071 [*]	2.157	.025	-13.64
DIVORCED	SINGLE	1.157	.991	.941	-1.86
	STEADY RELATIONSHIP	1.072	1.147	.983	-2.42
	LIVING WITH PARTNER	1.028	1.154	.987	-2.49
	MARRIED FIRST TIME	.161	.949	1.000	-2.73
	REMARRIED	-.333	1.208	1.000	-4.01
	SEPARATED	4.167	1.648	.186	-.85
	WIDOWED	-2.905	1.881	.783	-8.63
WIDOWED	SINGLE	4.062	1.709	.256	-1.15
	STEADY RELATIONSHIP	3.977	1.804	.351	-1.52
	LIVING WITH PARTNER	3.933	1.808	.369	-1.58
	MARRIED FIRST TIME	3.066	1.685	.607	-2.07
	REMARRIED	2.571	1.844	.859	-3.04
	SEPARATED	7.071 [*]	2.157	.025	.50
	DIVORCED	2.905	1.881	.783	-2.82

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) marital status	(J) marital status	95% ... Upper Bound
	LIVING WITH PARTNER	1.63
	MARRIED FIRST TIME	.32
	REMARRIED	.39
	DIVORCED	.85
	WIDOWED	-.50
DIVORCED	SINGLE	4.18
	STEADY RELATIONSHIP	4.57
	LIVING WITH PARTNER	4.54
	MARRIED FIRST TIME	3.05
	REMARRIED	3.35
	SEPARATED	9.19
	WIDOWED	2.82
WIDOWED	SINGLE	9.27
	STEADY RELATIONSHIP	9.47
	LIVING WITH PARTNER	9.44
	MARRIED FIRST TIME	8.20
	REMARRIED	8.19
	SEPARATED	13.64
	DIVORCED	8.63

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

Total Optimism

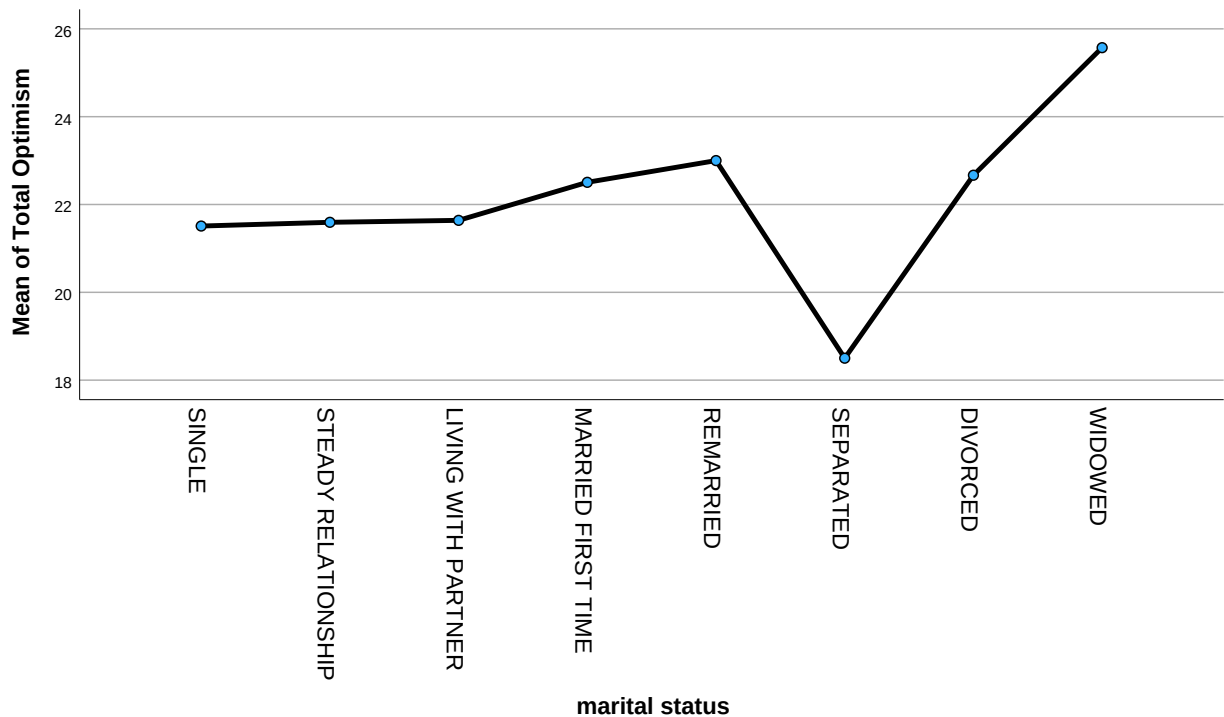
Tukey HSD^{a,b}

marital status	N	Subset for alpha = 0.05	
		1	2
SEPARATED	10	18.50	
SINGLE	104	21.51	21.51
STEADY RELATIONSHIP	37	21.59	21.59
LIVING WITH PARTNER	36	21.64	21.64
MARRIED FIRST TIME	188	22.51	22.51
DIVORCED	24		22.67
REMARRIED	29		23.00
WIDOWED	7		25.57
Sig.		.068	.061

Means for groups in homogeneous subsets are displayed.

- Uses Harmonic Mean Sample Size = 20.579.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not ...

Means Plots



Question3: Age Group and smoking on Optimism: Two-way Anova

Notes

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Between-Subjects Factors

		Value Label	N
age 3 groups	1	18 - 29	146
	2	30 - 44	153
	3	45+	133
smoker	1	YES	85
	2	NO	347

Descriptive Statistics

Dependent Variable: Total Optimism

age 3 groups	smoker	Mean	Std. Deviation	N
18 - 29	YES	20.50	5.550	34
	NO	21.63	4.217	112
	Total	21.36	4.567	146
30 - 44	YES	22.29	3.834	34
	NO	22.05	4.246	119
	Total	22.10	4.147	153
45+	YES	21.94	4.815	17
	NO	23.17	4.436	116
	Total	23.02	4.486	133
Total	YES	21.51	4.795	85
	NO	22.29	4.338	347
	Total	22.13	4.437	432

Levene's Test of Equality of Error Variances^{a,b}

		Levene Statistic	df1	df2	Sig.
Total Optimism	Based on Mean	2.085	5	426	.066
	Based on Median	1.459	5	426	.202
	Based on Median and with adjusted df	1.459	5	384.458	.202
	Based on trimmed mean	2.038	5	426	.072

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: Total Optimism

b. Design: Intercept + agegp3 + smoke + agegp3 * smoke

Tests of Between-Subjects Effects

Dependent Variable: Total Optimism

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	247.214 ^a	5	49.443	2.557	.027	.029
Intercept	120572.131	1	120572.131	6235.733	<.001	.936
agegp3	105.173	2	52.587	2.720	.067	.013
smoke	31.078	1	31.078	1.607	.206	.004
agegp3 * smoke	31.913	2	15.956	.825	.439	.004
Error	8236.999	426	19.336			
Total	220132.000	432				
Corrected Total	8484.213	431				

a. R Squared = .029 (Adjusted R Squared = .018)

Post Hoc Tests

age 3 groups

Multiple Comparisons

Dependent Variable: Total Optimism

Tukey HSD

(I) age 3 groups	(J) age 3 groups	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
18 - 29	30 - 44	-.74	.509	.313	-1.94	.45
	45+	-1.65*	.527	.005	-2.89	-.41
30 - 44	18 - 29	.74	.509	.313	-.45	1.94
	45+	-.91	.521	.189	-2.14	.32
45+	18 - 29	1.65*	.527	.005	.41	2.89
	30 - 44	.91	.521	.189	-.32	2.14

Based on observed means.

The error term is Mean Square(Error) = 19.336.

*. The mean difference is significant at the .05 level.

Homogeneous Subsets

Total Optimism

Tukey HSD^{a,b,c}

age 3 groups	N	Subset	
		1	2
18 - 29	146	21.36	
30 - 44	153	22.10	22.10
45+	133		23.02
Sig.		.327	.187

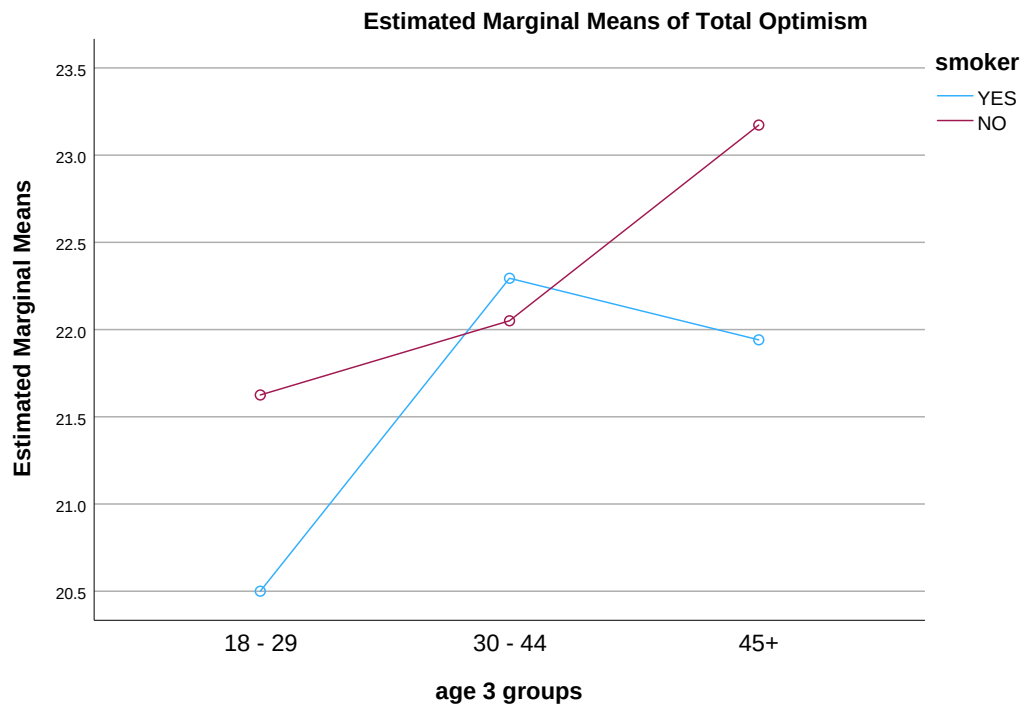
Means for groups in homogeneous subsets are displayed.

Based on observed means.

The error term is Mean Square(Error) = 19.336.

- Uses Harmonic Mean Sample Size = 143.513.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.
- Alpha = .05.

Profile Plots



Explore

Notes

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	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.
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married & remarried

Case Processing Summary

		Cases				
		Valid		Missing		Total
	married & remarried	N	Percent	N	Percent	N
Total life satisfaction	1.00	218	99.5%	1	0.5%	219
	2.00	218	99.1%	2	0.9%	220

Case Processing Summary

		Cases
		Total
	married & remarried	Percent
Total life satisfaction	1.00	100.0%
	2.00	100.0%

Descriptives

		married & remarried	Statistic	Std. Error
Total life satisfaction	1.00	Mean	23.57	.452
		95% Confidence Interval for Mean	Lower Bound	22.68
			Upper Bound	24.46
		5% Trimmed Mean	23.80	
		Median	24.00	
		Variance	44.569	
		Std. Deviation	6.676	
		Minimum	5	
		Maximum	35	
		Range	30	
		Interquartile Range	10	
		Skewness	-.512	.165
		Kurtosis	-.338	.328
	2.00	Mean	21.19	.452
		95% Confidence Interval for Mean	Lower Bound	20.30
			Upper Bound	22.08
		5% Trimmed Mean	21.25	
		Median	22.00	
		Variance	44.448	
		Std. Deviation	6.667	
		Minimum	5	
		Maximum	35	
		Range	30	
		Interquartile Range	9	
		Skewness	-.168	.165
		Kurtosis	-.351	.328

Tests of Normality

	married & remarried	Kolmogorov-Smirnov ^a			Shapiro-Wilk	
		Statistic	df	Sig.	Statistic	df
Total life satisfaction	1.00	.094	218	<.001	.968	218
	2.00	.080	218	.002	.988	218

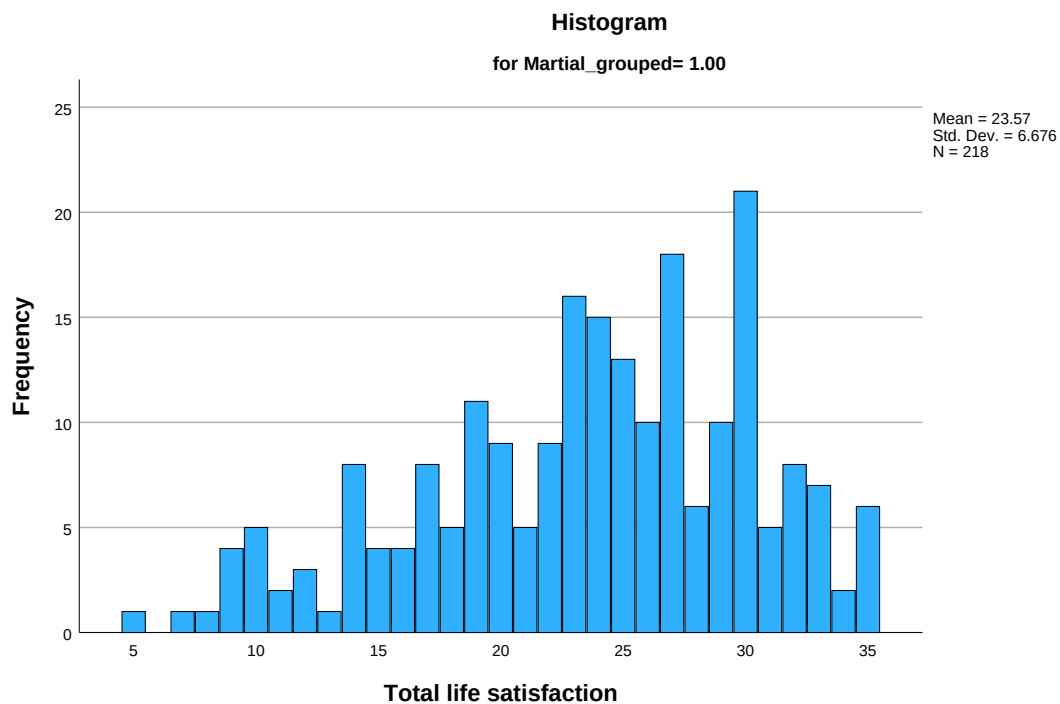
Tests of Normality

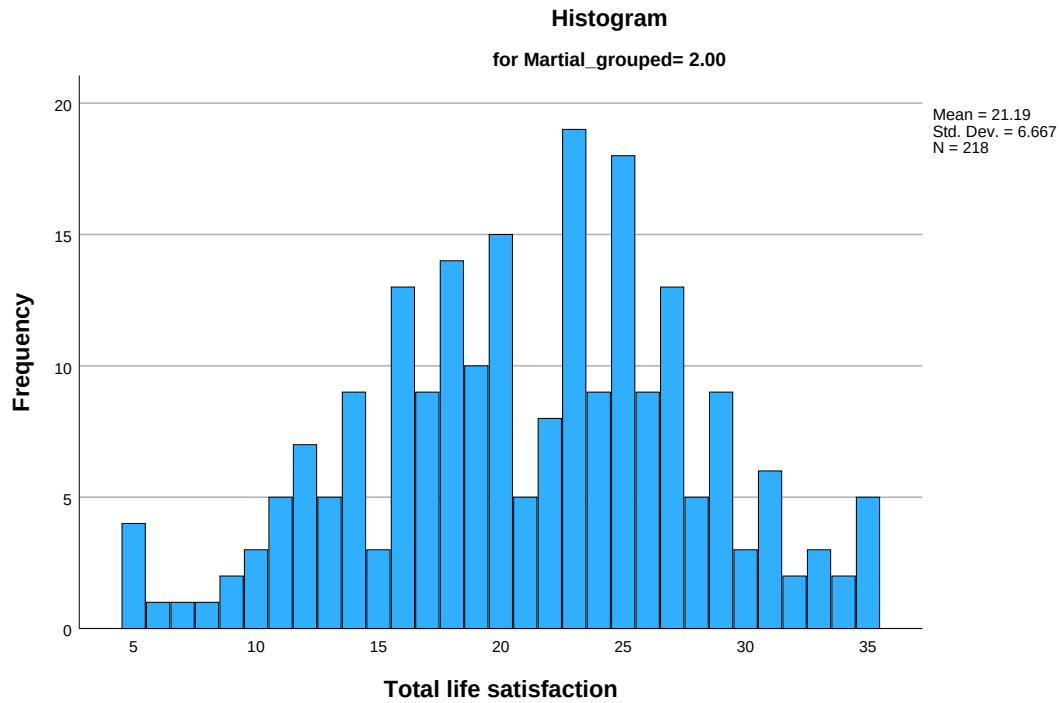
	married & remarried	Shapiro-...
		Sig.
Total life satisfaction	1.00	<.001
	2.00	.073

a. Lilliefors Significance Correction

Total life satisfaction

Histograms





Stem-and-Leaf Plots

Total Life satisfaction Stem-and-Leaf Plot for Martial_grouped= 1.00

Frequency	Stem & Leaf
1.00	0 . 5
1.00	0 . 7
5.00	0 . 89999
7.00	1 . 0000011
4.00	1 . 2223
12.00	1 . 444444445555
12.00	1 . 666677777777
16.00	1 . 8888899999999999
14.00	2 . 0000000011111
25.00	2 . 222222223333333333333333
28.00	2 . 444444444444444455555555555555
28.00	2 . 666666666666777777777777777777
16.00	2 . 8888899999999999
26.00	3 . 00000000000000000000000011111
15.00	3 . 22222223333333
8.00	3 . 4455555

Stem width: 10

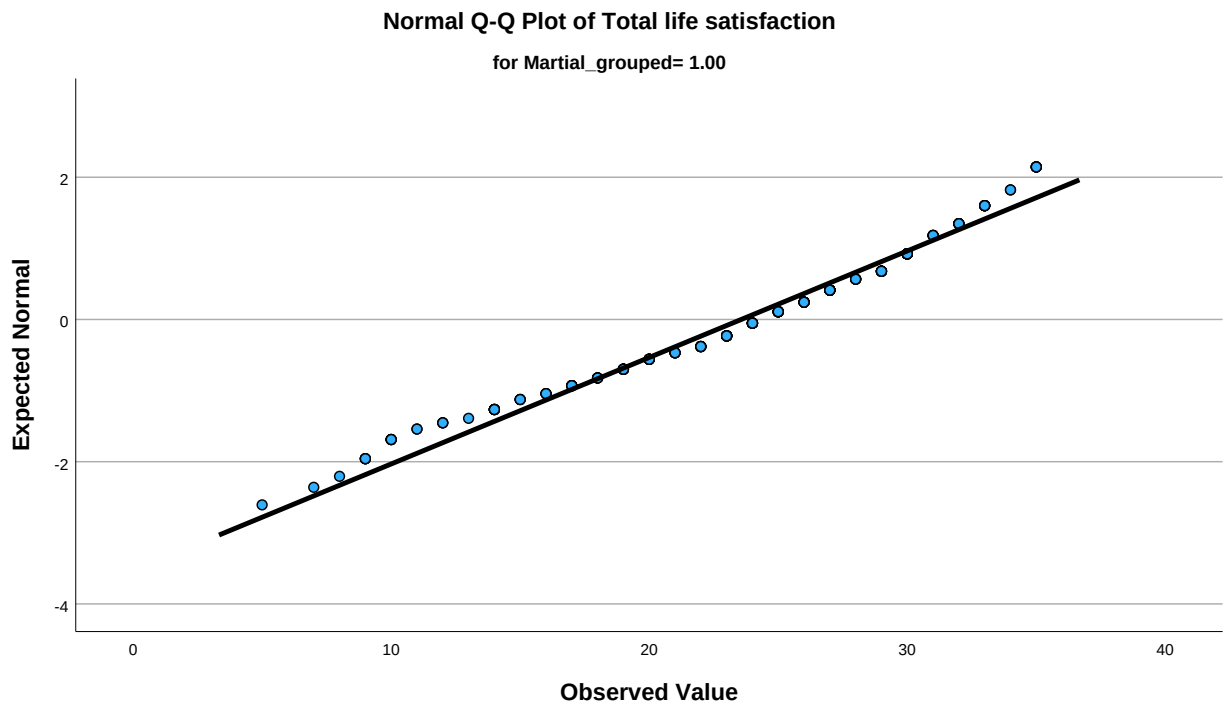
Each leaf: 1 case(s)

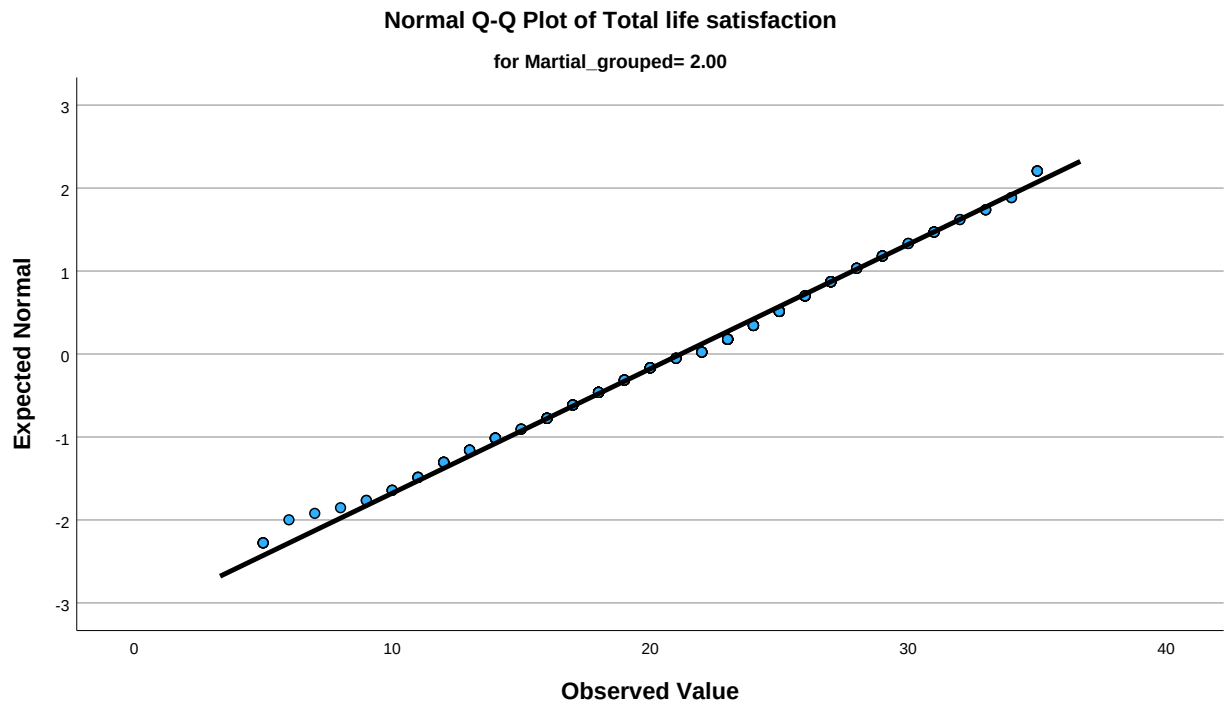
Total Life satisfaction Stem-and-Leaf Plot for
 Martial_grouped= 2.00

Frequency	Stem & Leaf
4.00	0 . 5555
2.00	0 . 67
3.00	0 . 899
8.00	1 . 00011111
12.00	1 . 22222233333
12.00	1 . 44444444555
22.00	1 . 66666666666677777777
24.00	1 . 88888888888888999999999
20.00	2 . 0000000000000011111
27.00	2 . 222222233333333333333333333
27.00	2 . 444444445555555555555555555
22.00	2 . 66666666777777777777
14.00	2 . 8888999999999
9.00	3 . 000111111
5.00	3 . 22333
7.00	3 . 445555

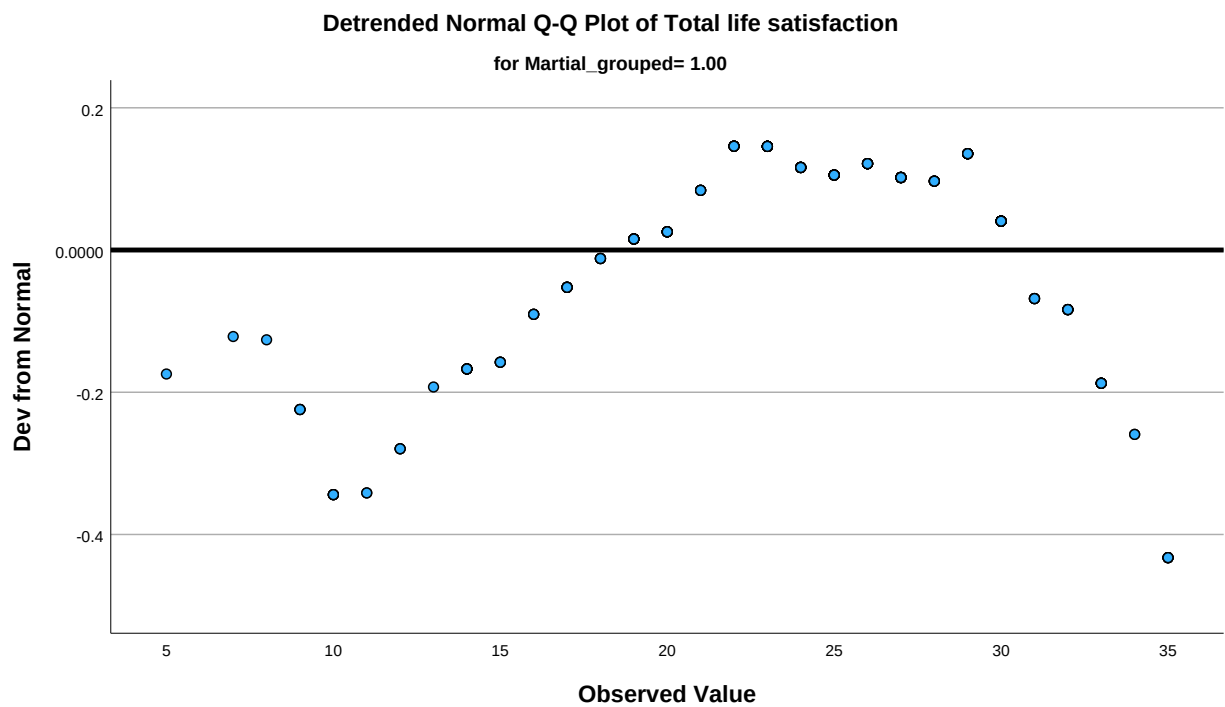
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 Each leaf: 1 case(s)

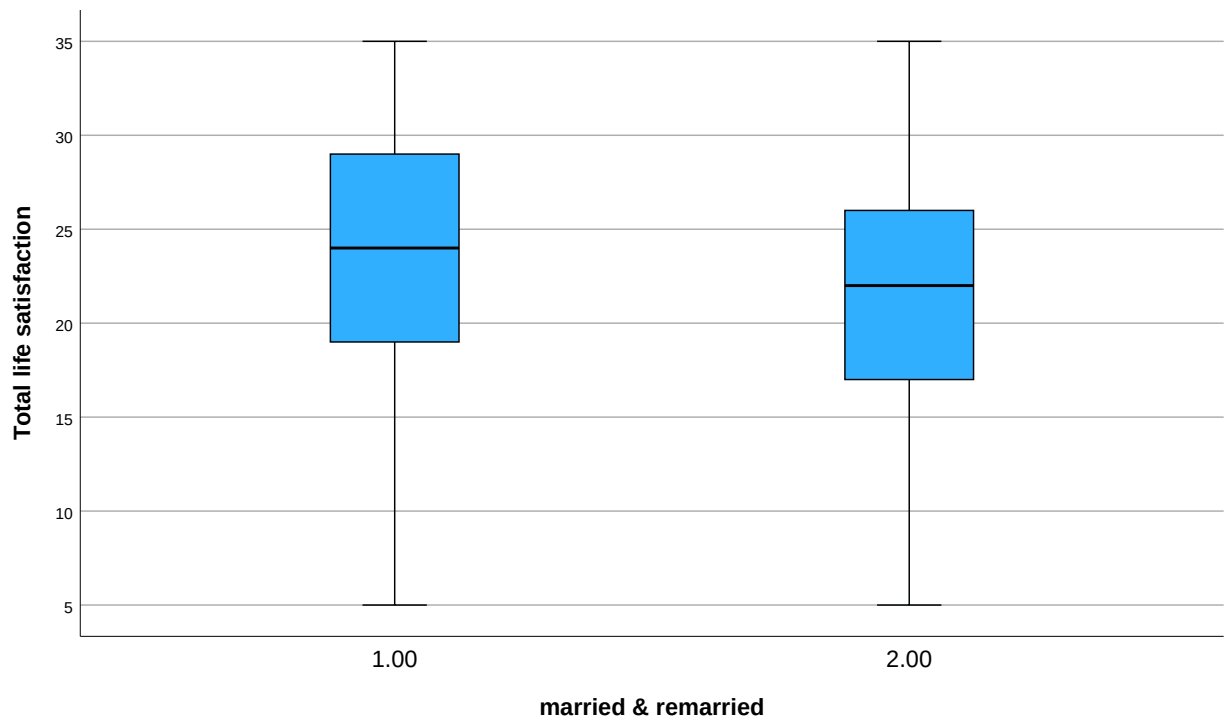
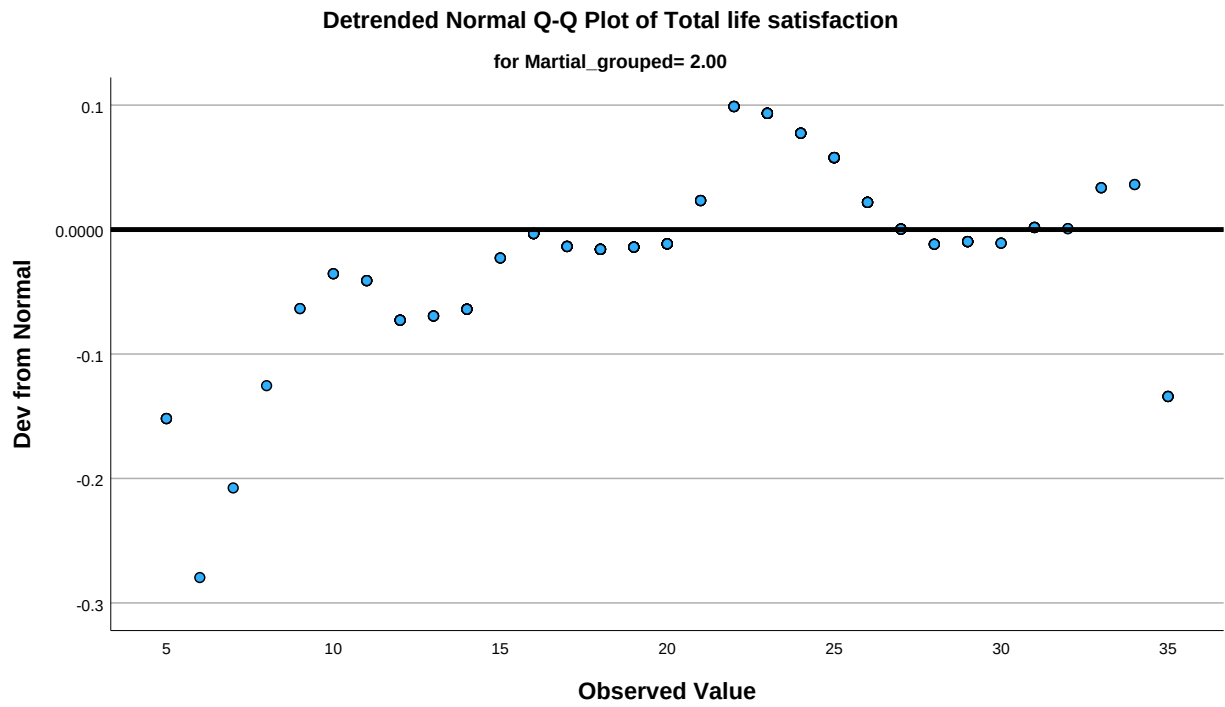
Normal Q-Q Plots





Detrended Normal Q-Q Plots





Explore

Notes

Output Created		06-NOV-2024 16:20:09
Comments		
Input	Data	D:\survey.sav
	Active Dataset	DataSet2
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	439
Missing Value Handling	Definition of Missing	User-defined missing values for dependent variables are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.
Syntax		EXAMINE VARIABLES=toptim BY marital /PLOT BOXPLOT STEMLEAF HISTOGRAM NPLOT /COMPARE GROUPS /STATISTICS DESCRIPTIVES /INTERVAL 95 /MISSING LISTWISE /NOTOTAL.
Resources	Processor Time	00:00:02.88
	Elapsed Time	00:00:02.24

Normality check for Optimism by Marital Status

Case Processing Summary

		Cases				
		Valid		Missing		Total
	marital status	N	Percent	N	Percent	N
Total Optimism	SINGLE	104	99.0%	1	1.0%	105
	STEADY RELATIONSHIP	37	100.0%	0	0.0%	37
	LIVING WITH PARTNER	36	97.3%	1	2.7%	37
	MARRIED FIRST TIME	188	99.5%	1	0.5%	189
	REMARRIED	29	96.7%	1	3.3%	30
	SEPARATED	10	100.0%	0	0.0%	10
	DIVORCED	24	100.0%	0	0.0%	24
	WIDOWED	7	100.0%	0	0.0%	7

Case Processing Summary

		Cases
		Total
	marital status	Percent
Total Optimism	SINGLE	100.0%
	STEADY RELATIONSHIP	100.0%
	LIVING WITH PARTNER	100.0%
	MARRIED FIRST TIME	100.0%
	REMARRIED	100.0%
	SEPARATED	100.0%
	DIVORCED	100.0%
	WIDOWED	100.0%

Descriptives

marital status		Statistic	
Total Optimism	SINGLE	Mean	21.51
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	21.65
		Median	21.00
		Variance	20.427
		Std. Deviation	4.520
		Minimum	7
		Maximum	30
		Range	23
		Interquartile Range	6
		Skewness	-.450
		Kurtosis	.532
	STEADY RELATIONSHIP	Mean	21.59
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	21.75
		Median	22.00
		Variance	20.470
		Std. Deviation	4.524
		Minimum	10
		Maximum	29
		Range	19
		Interquartile Range	6
		Skewness	-.538
		Kurtosis	.011
	LIVING WITH PARTNER	Mean	21.64
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	21.96
		Median	22.50
		Variance	20.694
		Std. Deviation	4.549
		Minimum	8
		Maximum	29
		Range	21
		Interquartile Range	4
		Skewness	-1.151
		Kurtosis	1.903

Descriptives

marital status				Std. Error
Total Optimism	SINGLE	Mean		.443
		95% Confidence Interval for Mean	Lower Bound	
			Upper Bound	
		5% Trimmed Mean		
		Median		
		Variance		
		Std. Deviation		
		Minimum		
		Maximum		
		Range		
		Interquartile Range		
		Skewness		.237
		Kurtosis		.469
	STEADY RELATIONSHIP	Mean		.744
		95% Confidence Interval for Mean	Lower Bound	
			Upper Bound	
		5% Trimmed Mean		
		Median		
		Variance		
		Std. Deviation		
		Minimum		
		Maximum		
		Range		
		Interquartile Range		
		Skewness		.388
		Kurtosis		.759
	LIVING WITH PARTNER	Mean		.758
		95% Confidence Interval for Mean	Lower Bound	
			Upper Bound	
		5% Trimmed Mean		
		Median		
		Variance		
		Std. Deviation		
		Minimum		
		Maximum		
		Range		
		Interquartile Range		
		Skewness		.393
		Kurtosis		.768

Descriptives

marital status		Statistic
MARRIED FIRST TIME	Mean	22.51
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	22.64
	Median	23.00
	Variance	17.631
	Std. Deviation	4.199
	Minimum	9
	Maximum	30
	Range	21
	Interquartile Range	6
	Skewness	-.436
	Kurtosis	.282
REMARRIED	Mean	23.00
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	23.11
	Median	24.00
	Variance	20.143
	Std. Deviation	4.488
	Minimum	15
	Maximum	29
	Range	14
	Interquartile Range	9
	Skewness	-.491
	Kurtosis	-1.261
SEPARATED	Mean	18.50
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	18.44
	Median	18.00
	Variance	24.944
	Std. Deviation	4.994
	Minimum	10
	Maximum	28
	Range	18
	Interquartile Range	7
	Skewness	.298
	Kurtosis	.705

Descriptives

marital status		Std. Error
MARRIED FIRST TIME	Mean	.306
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	
	Median	
	Variance	
	Std. Deviation	
	Minimum	
	Maximum	
	Range	
	Interquartile Range	
	Skewness	.177
	Kurtosis	.353
REMARRIED	Mean	.833
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	
	Median	
	Variance	
	Std. Deviation	
	Minimum	
	Maximum	
	Range	
	Interquartile Range	
	Skewness	.434
	Kurtosis	.845
SEPARATED	Mean	1.579
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	
	Median	
	Variance	
	Std. Deviation	
	Minimum	
	Maximum	
	Range	
	Interquartile Range	
	Skewness	.687
	Kurtosis	1.334

Descriptives

marital status			Statistic	
DIVORCED	Mean		22.67	
	95% Confidence Interval for Mean	Lower Bound	20.86	
		Upper Bound	24.47	
	5% Trimmed Mean		22.68	
	Median		23.00	
	Variance		18.319	
	Std. Deviation		4.280	
	Minimum		15	
	Maximum		30	
	Range		15	
	Interquartile Range		8	
	Skewness		.040	
	Kurtosis		-.840	
	WIDOWED	Mean		25.57
		95% Confidence Interval for Mean	Lower Bound	21.58
Upper Bound			29.56	
5% Trimmed Mean		25.91		
Median		27.00		
Variance		18.619		
Std. Deviation		4.315		
Minimum		16		
Maximum		29		
Range		13		
Interquartile Range		1		
Skewness		-2.396		
Kurtosis		6.106		

Descriptives

marital status		Std. Error
DIVORCED	Mean	.874
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	
	Median	
	Variance	
	Std. Deviation	
	Minimum	
	Maximum	
	Range	
	Interquartile Range	
	Skewness	.472
	Kurtosis	.918
WIDOWED	Mean	1.631
	95% Confidence Interval for Mean	Lower Bound
		Upper Bound
	5% Trimmed Mean	
	Median	
	Variance	
	Std. Deviation	
	Minimum	
	Maximum	
	Range	
	Interquartile Range	
	Skewness	.794
	Kurtosis	1.587

Tests of Normality

	marital status	Kolmogorov-Smirnov ^a			Shapiro-Wilk	
		Statistic	df	Sig.	Statistic	df
Total Optimism	SINGLE	.064	104	.200 [*]	.976	104
	STEADY RELATIONSHIP	.108	37	.200 [*]	.971	37
	LIVING WITH PARTNER	.166	36	.013	.915	36
	MARRIED FIRST TIME	.070	188	.026	.976	188
	REMARRIED	.196	29	.006	.883	29
	SEPARATED	.160	10	.200 [*]	.978	10
	DIVORCED	.112	24	.200 [*]	.964	24
	WIDOWED	.397	7	.001	.632	7

Tests of Normality

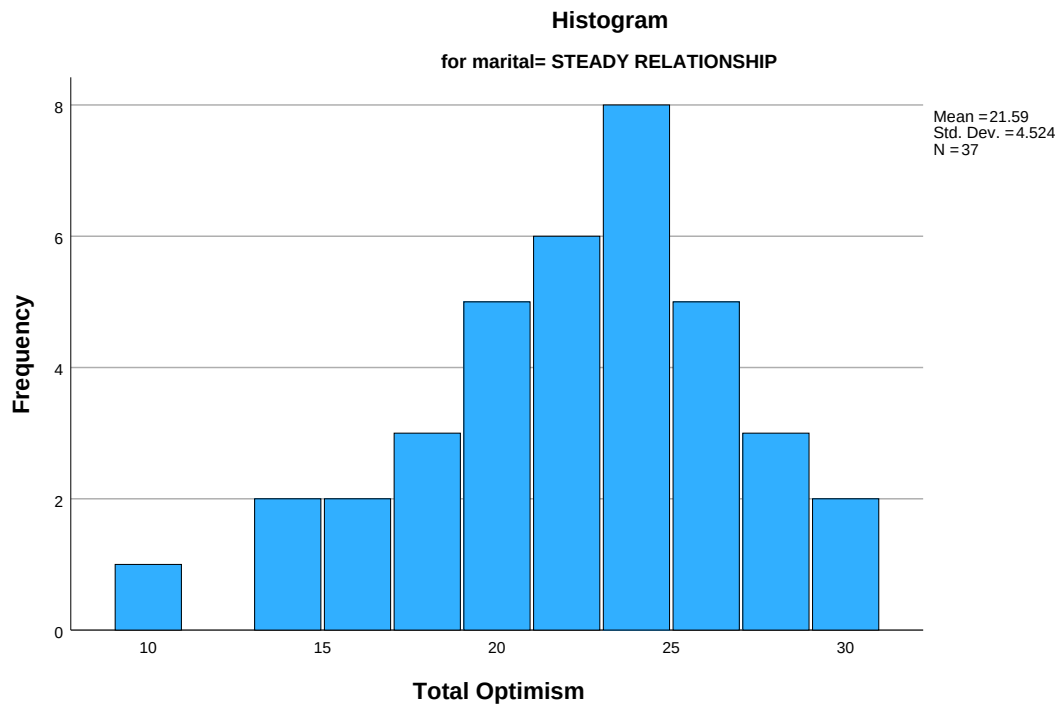
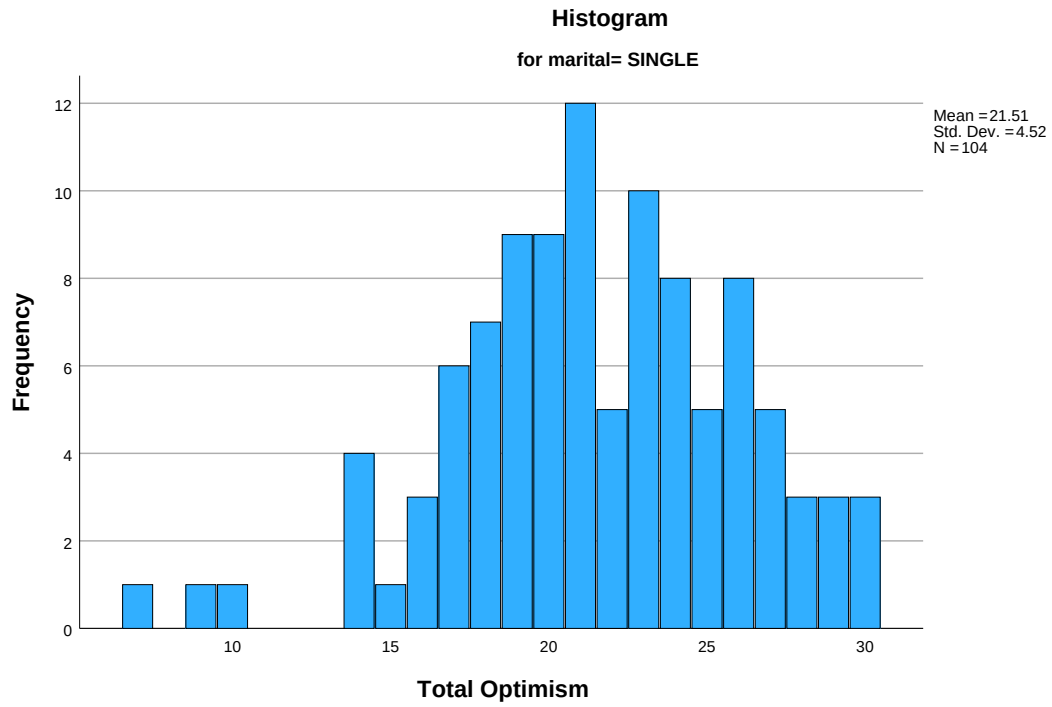
	marital status	Shapiro-...
		Sig.
Total Optimism	SINGLE	.056
	STEADY RELATIONSHIP	.428
	LIVING WITH PARTNER	.009
	MARRIED FIRST TIME	.003
	REMARRIED	.004
	SEPARATED	.955
	DIVORCED	.529
	WIDOWED	<.001

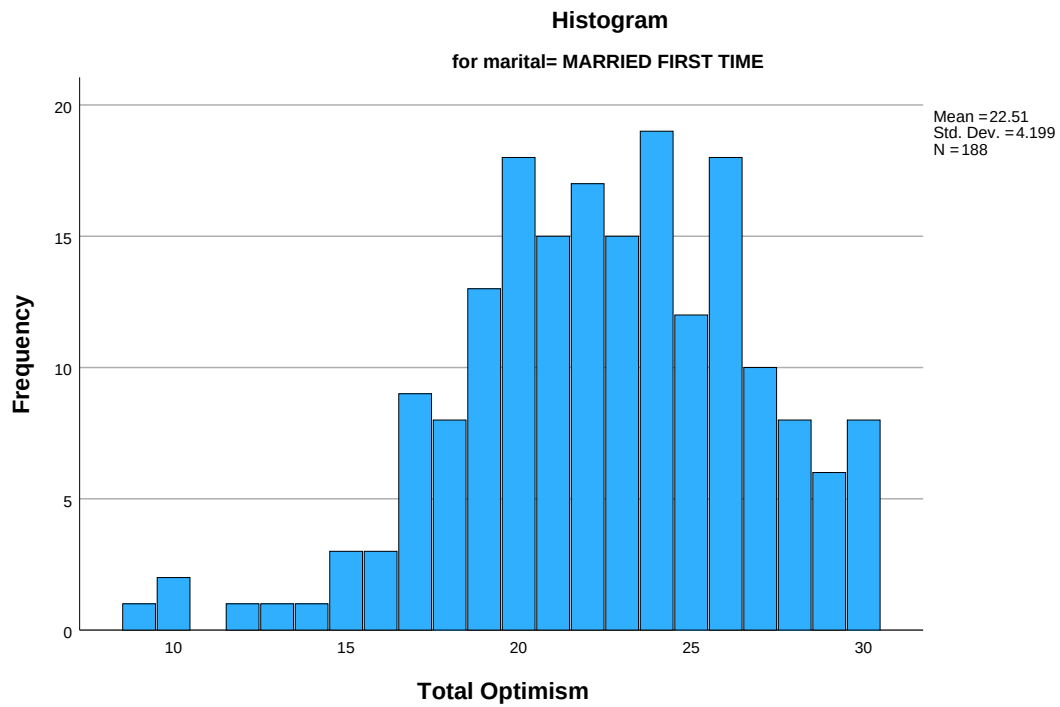
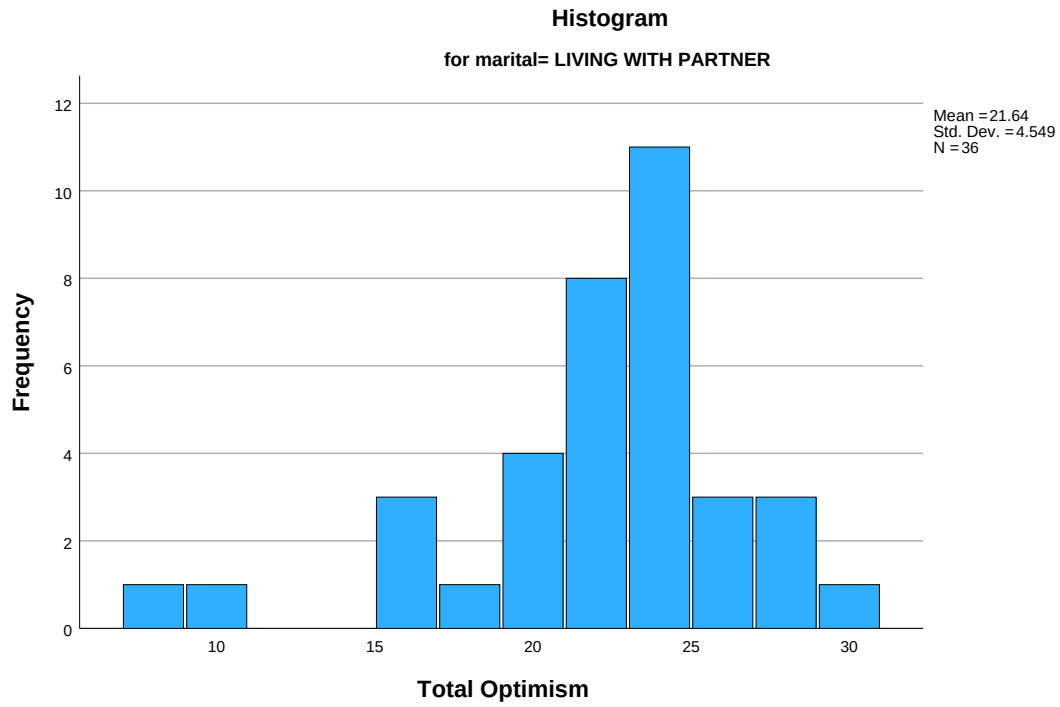
*. This is a lower bound of the true significance.

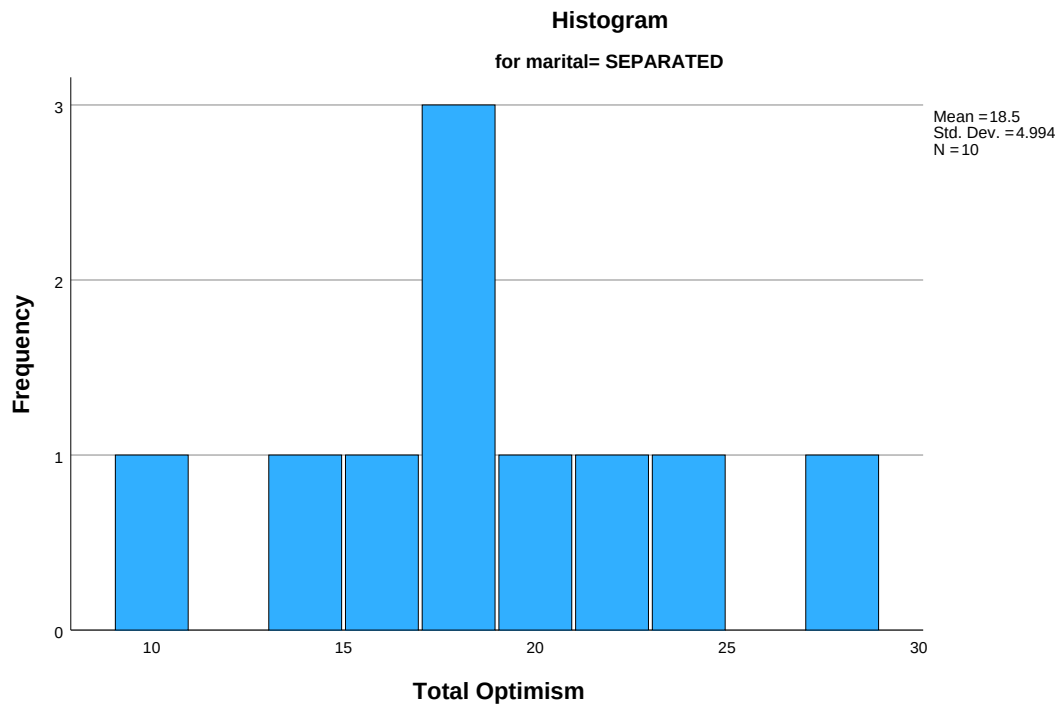
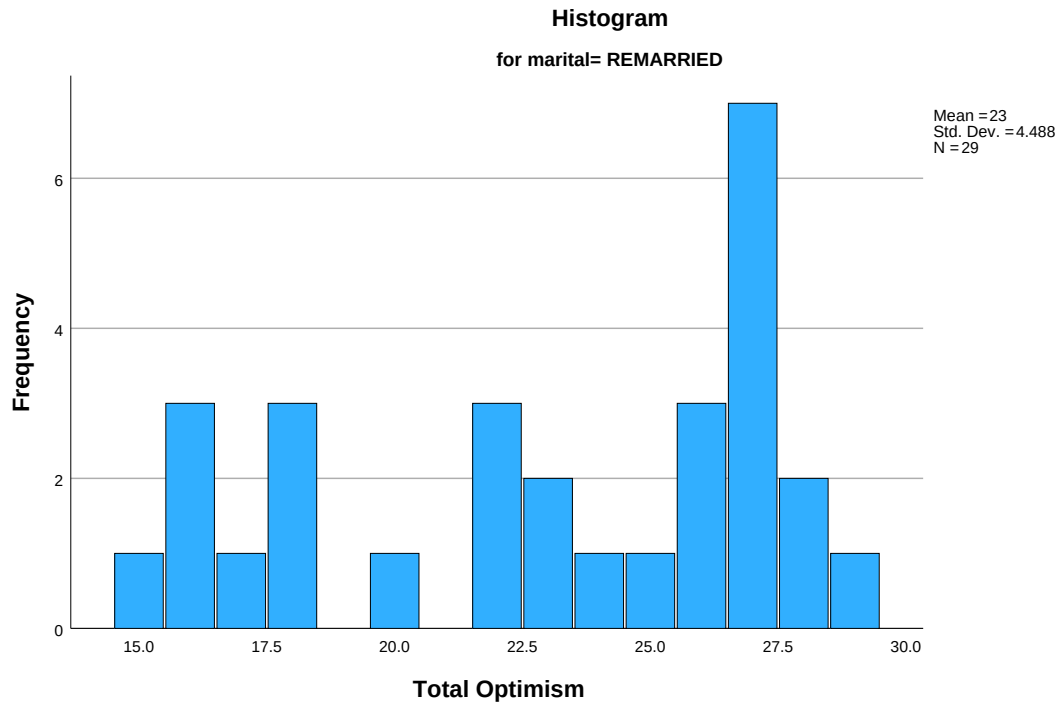
a. Lilliefors Significance Correction

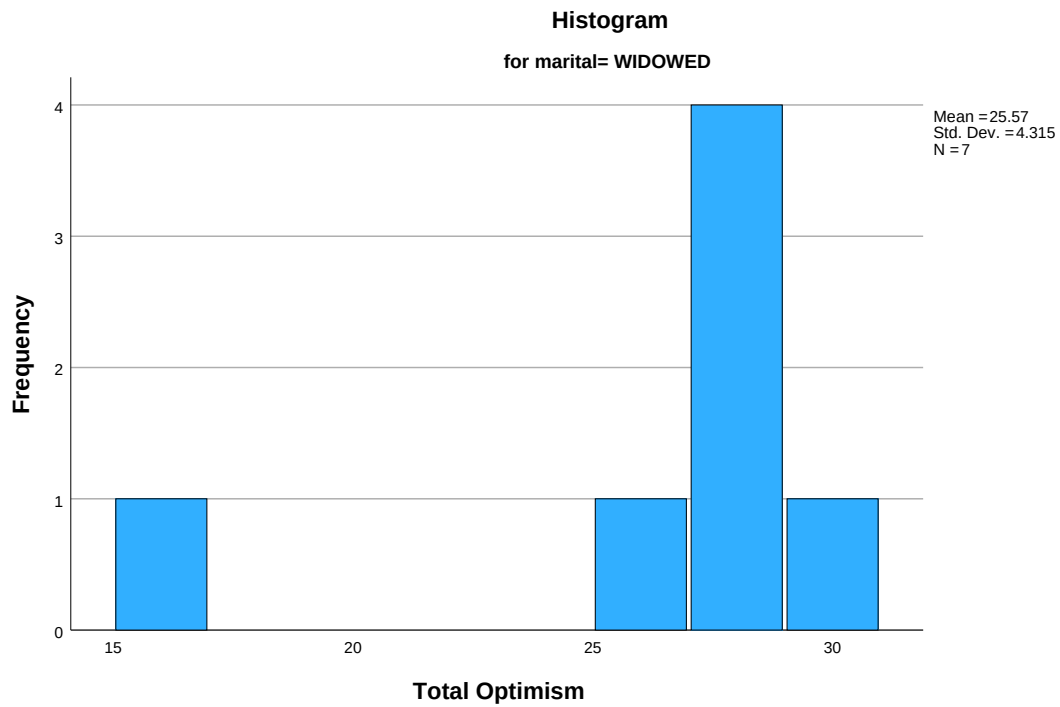
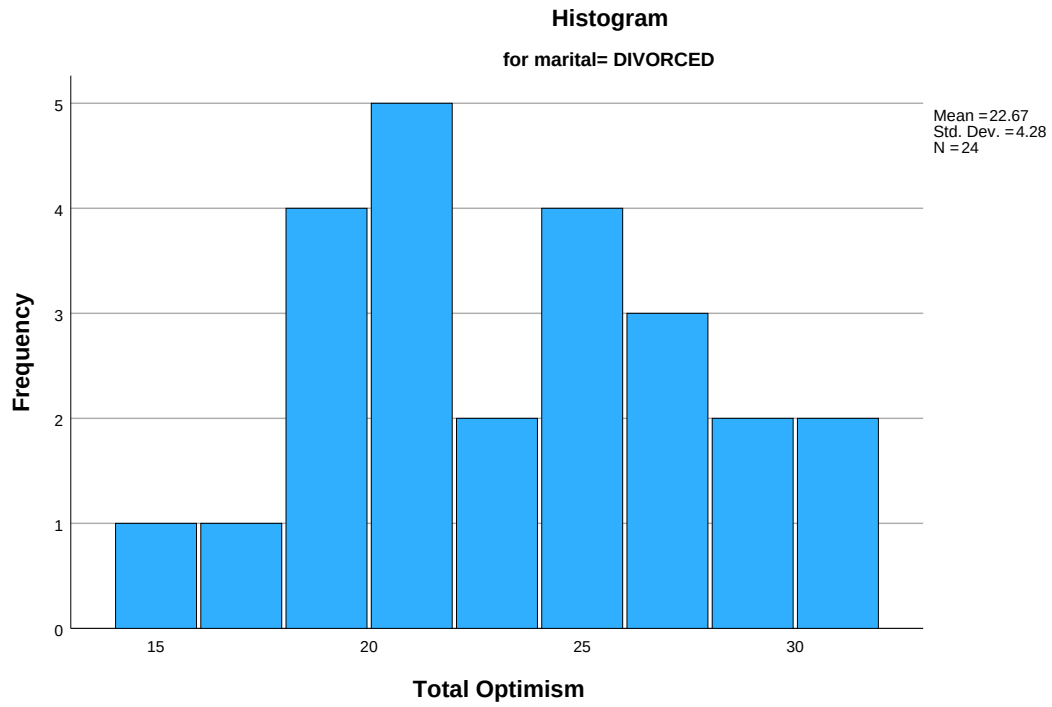
Total Optimism

Histograms









Stem-and-Leaf Plots

Total Optimism Stem-and-Leaf Plot for
marital= SINGLE

Frequency Stem & Leaf

2.00 Extremes (<=9)
 1.00 1 . 0
 .00 1 .
 5.00 1 . 4445
 9.00 1 . 66677777
 16.00 1 . 888888999999999
 21.00 2 . 00000000111111111111
 15.00 2 . 22223333333333
 13.00 2 . 444444455555
 13.00 2 . 666666677777
 6.00 2 . 88899
 3.00 3 . 000

Stem width: 10
 Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
 marital= STEADY RELATIONSHIP

Frequency Stem & Leaf

1.00 1 . 0
 1.00 1 . 3
 2.00 1 . 45
 4.00 1 . 6777
 2.00 1 . 99
 7.00 2 . 0001111
 7.00 2 . 2233333
 6.00 2 . 444555
 4.00 2 . 6677
 3.00 2 . 800

Stem width: 10
 Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
 marital= LIVING WITH PARTNER

Frequency Stem & Leaf

2.00 Extremes (<=10)
 1.00 1 . 5
 3.00 1 . 667
 2.00 1 . 99
 7.00 2 . 0011111
 7.00 2 . 2223333
 9.00 2 . 44444455

2.00	2 . 67
3.00	2 . 889

Stem width: 10
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
marital= MARRIED FIRST TIME

Frequency Stem & Leaf

3.00	Extremes	(=<10.0)
1.00	12 .	0
1.00	13 .	0
1.00	14 .	0
3.00	15 .	000
3.00	16 .	000
9.00	17 .	000000000
8.00	18 .	00000000
13.00	19 .	000000000000
18.00	20 .	0000000000000000
15.00	21 .	00000000000000
17.00	22 .	0000000000000000
15.00	23 .	00000000000000
19.00	24 .	0000000000000000
12.00	25 .	0000000000
18.00	26 .	0000000000000000
10.00	27 .	000000000
8.00	28 .	00000000
6.00	29 .	000000
8.00	30 .	00000000

Stem width: 1
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
marital= REMARRIED

Frequency Stem & Leaf

1.00	1 .	5
4.00	1 .	6667
3.00	1 .	888
1.00	2 .	0
5.00	2 .	22233
2.00	2 .	45
10.00	2 .	6667777777
3.00	2 .	889

Stem width: 10
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
marital= SEPARATED

Frequency Stem & Leaf

2.00	1 . 04
5.00	1 . 67889
2.00	2 . 23
1.00	2 . 8

Stem width: 10
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
marital= DIVORCED

Frequency Stem & Leaf

.00	1 .
6.00	1 . 568888
11.00	2 . 01111334444
5.00	2 . 66788
2.00	3 . 00

Stem width: 10
Each leaf: 1 case(s)

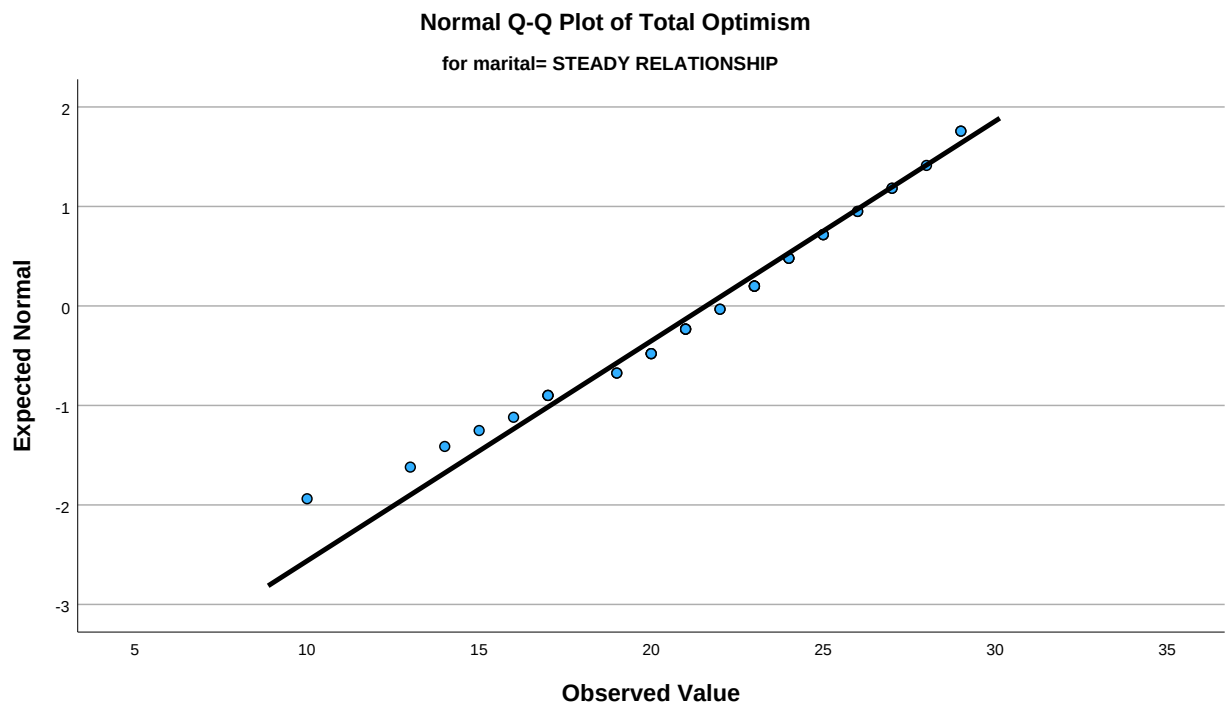
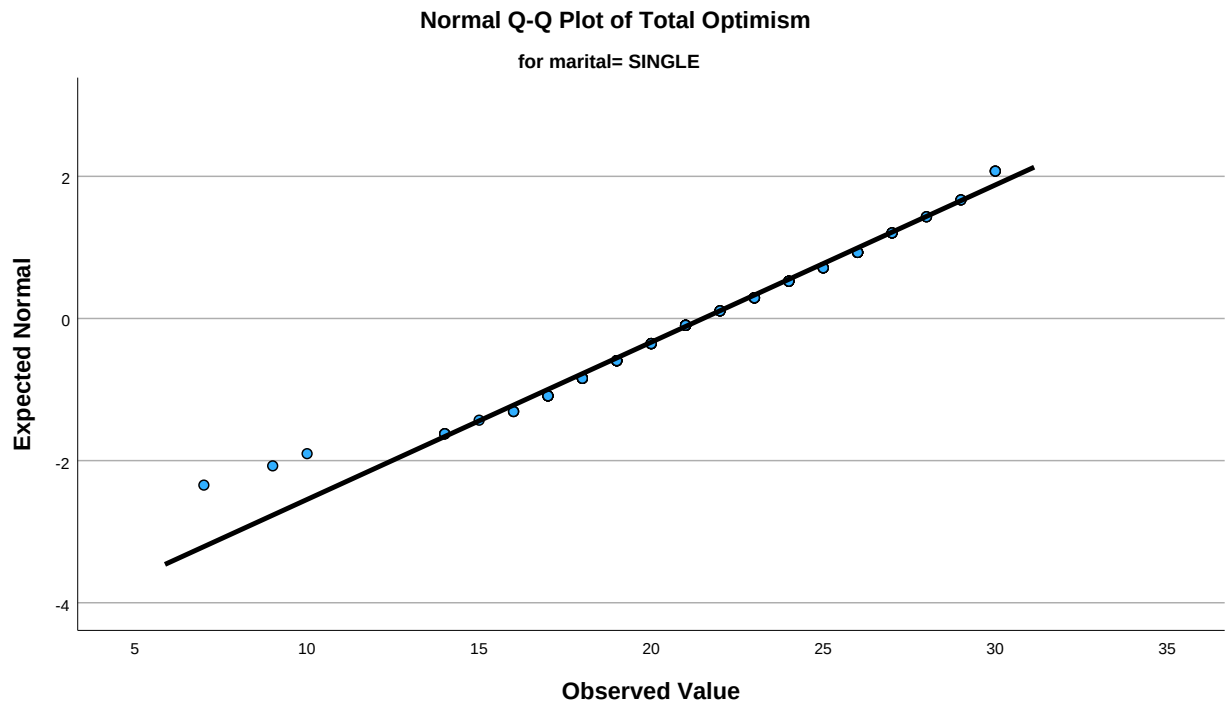
Total Optimism Stem-and-Leaf Plot for
marital= WIDOWED

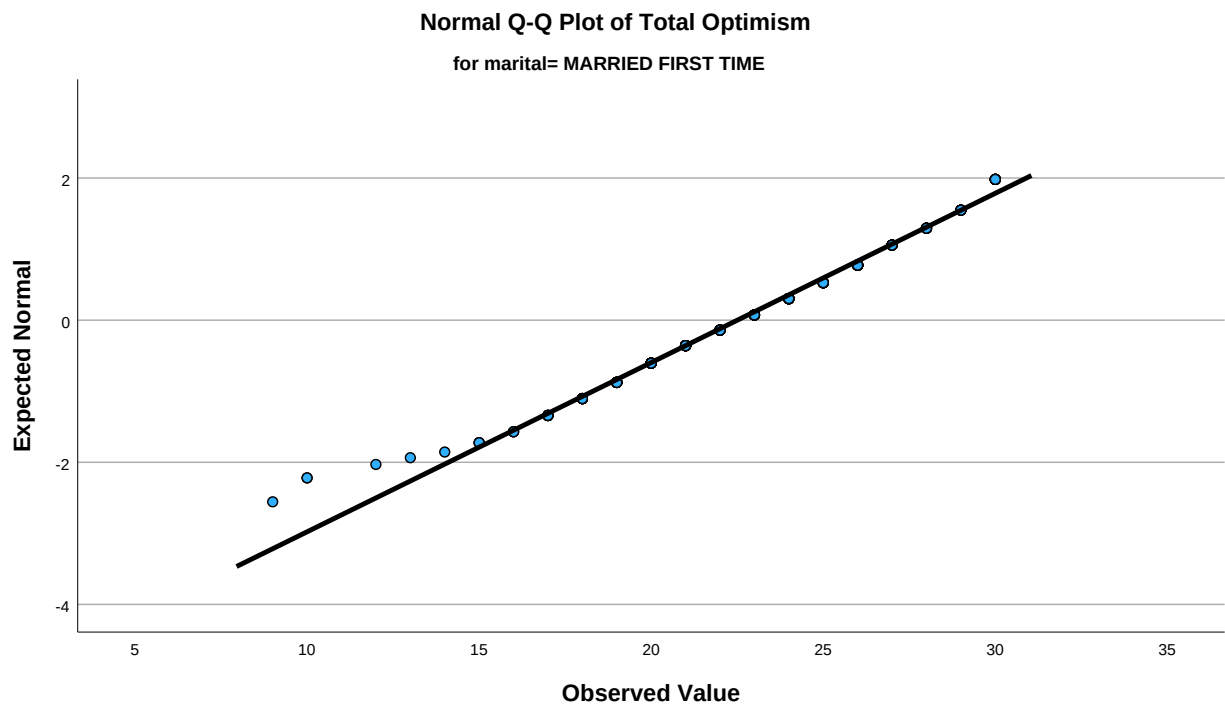
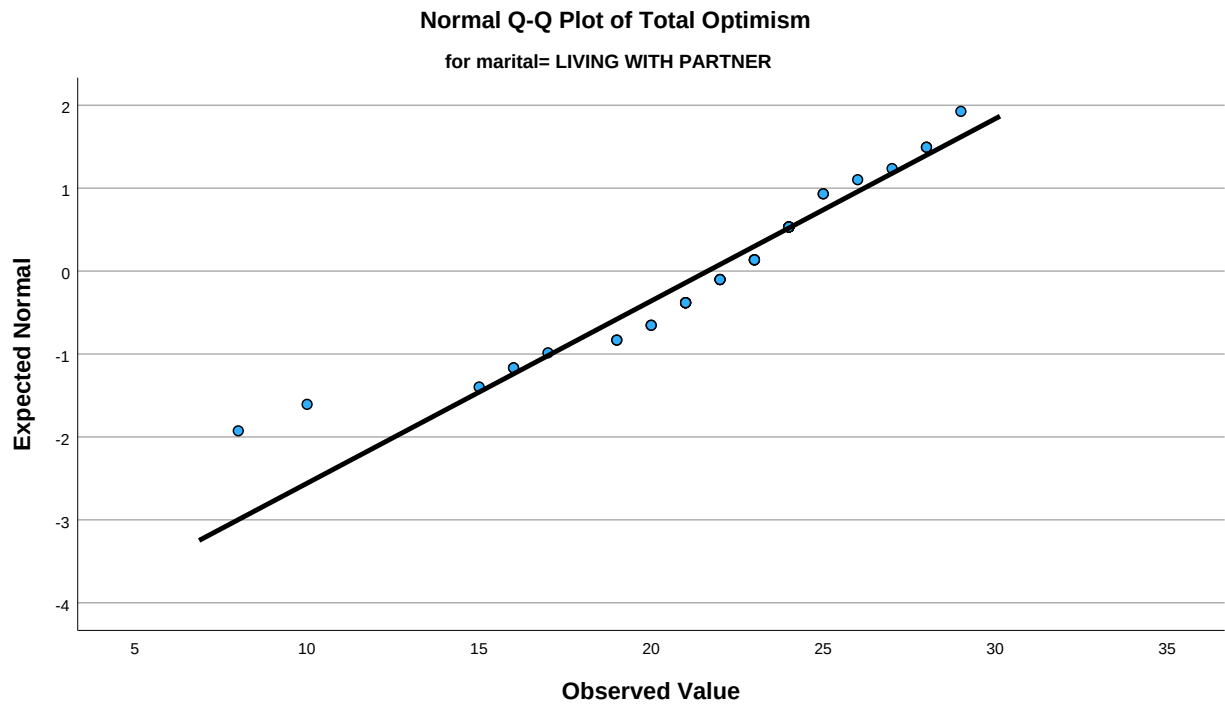
Frequency Stem & Leaf

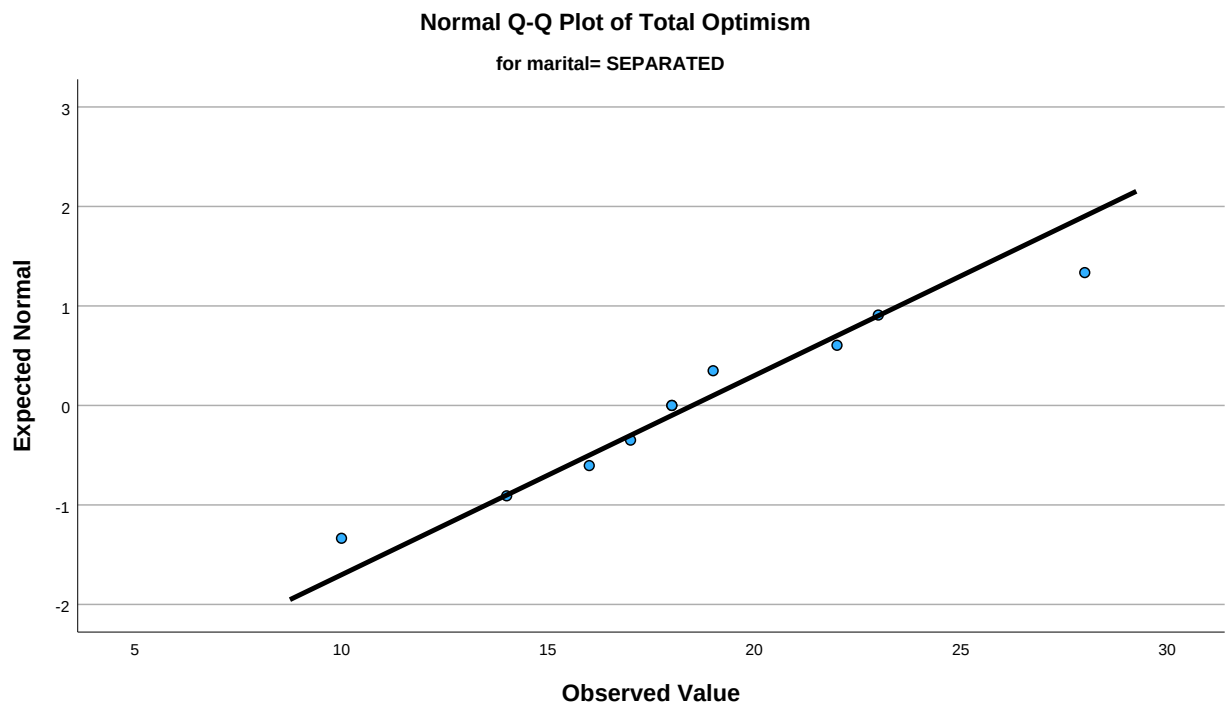
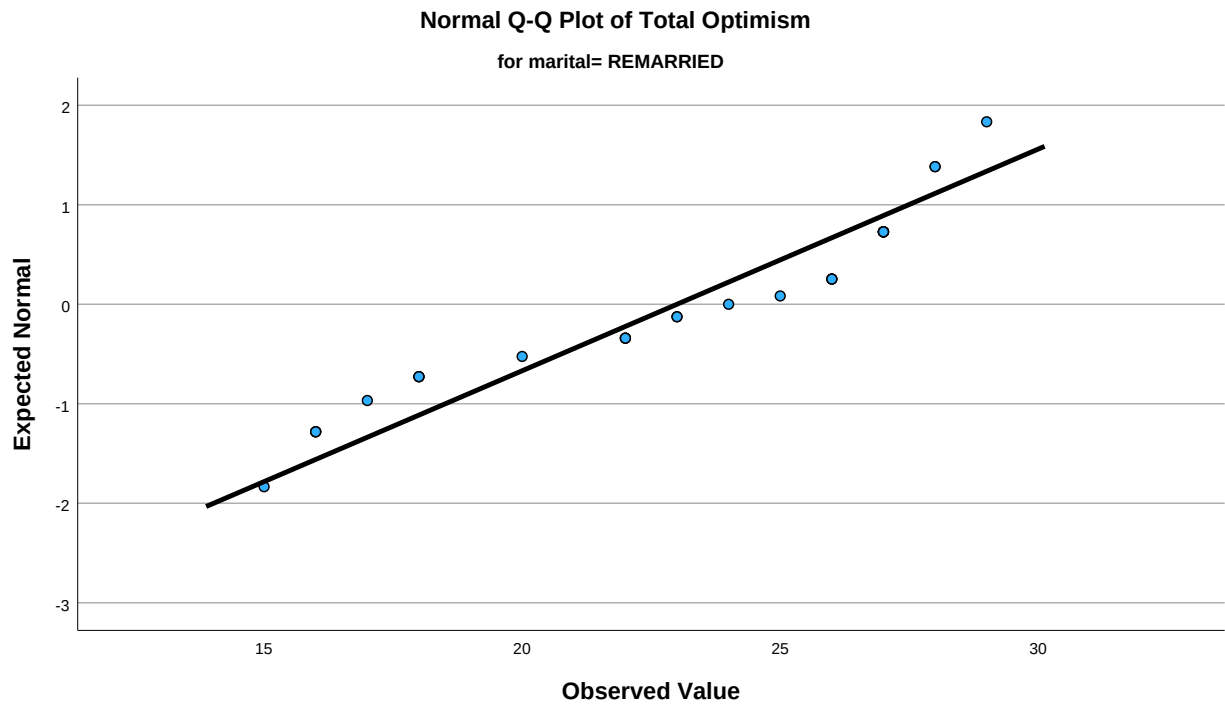
1.00	Extremes	(=<16.0)
1.00	26 . 0	
.00	26 .	
4.00	27 . 0000	
1.00	Extremes	(>=29.0)

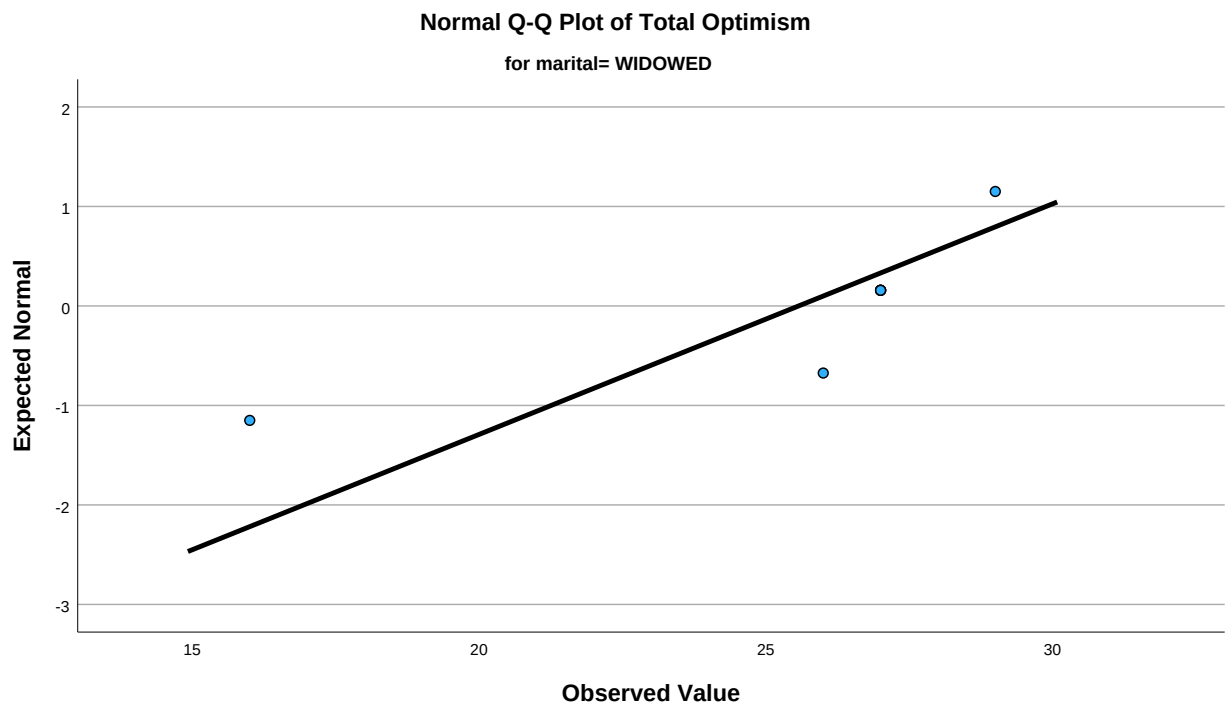
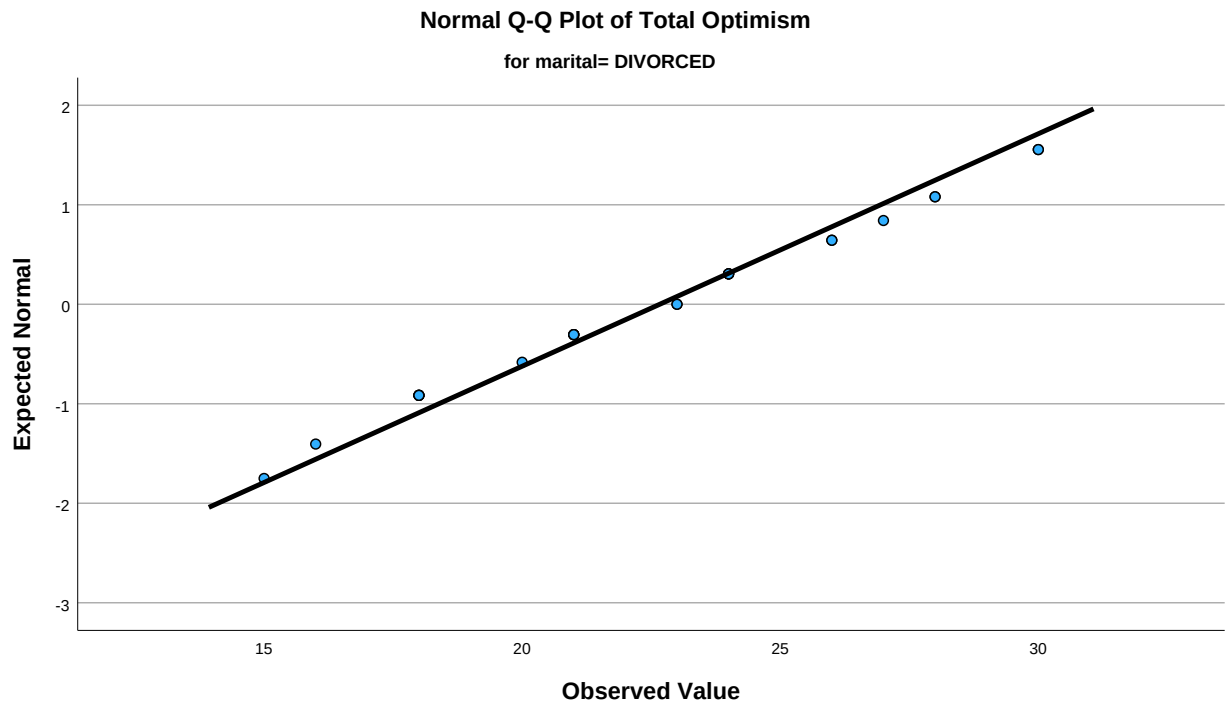
Stem width: 1
Each leaf: 1 case(s)

Normal Q-Q Plots

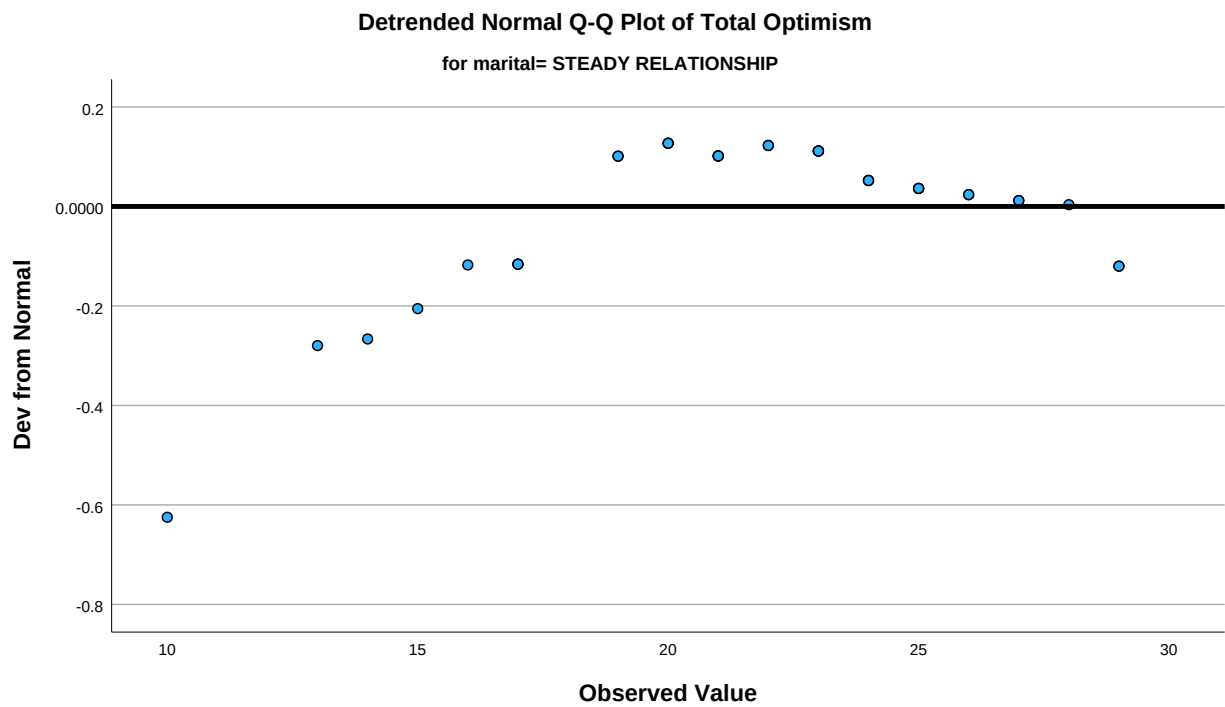
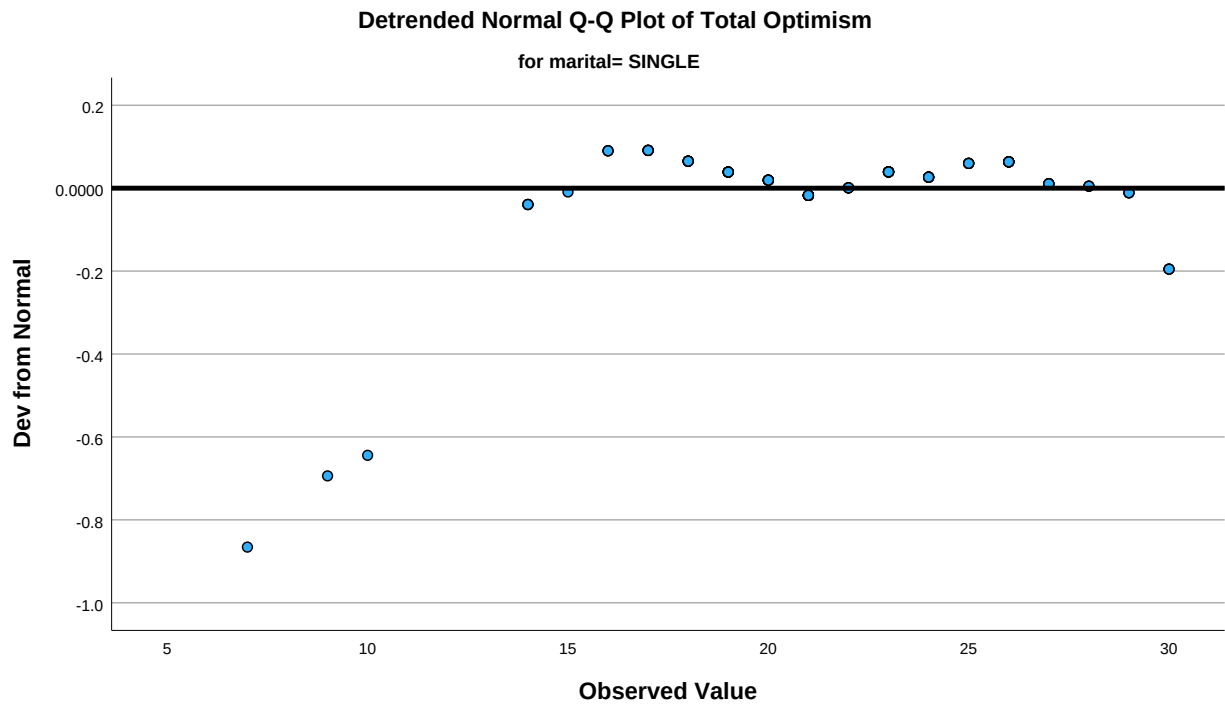


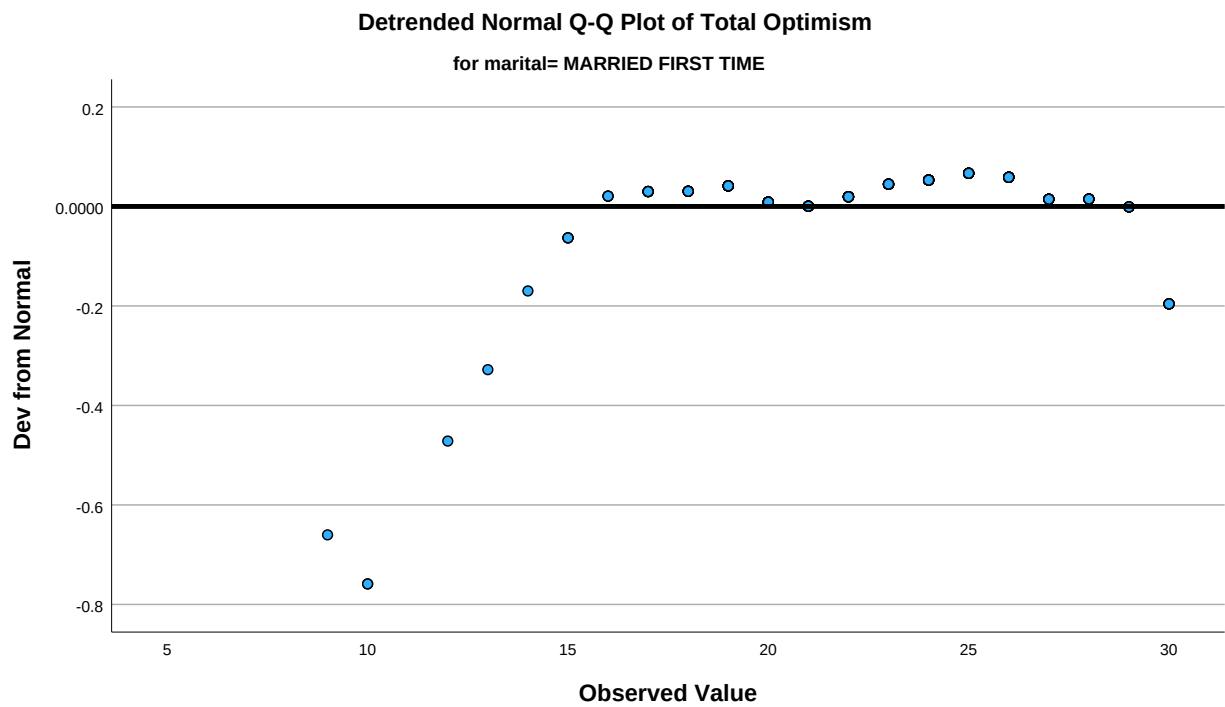
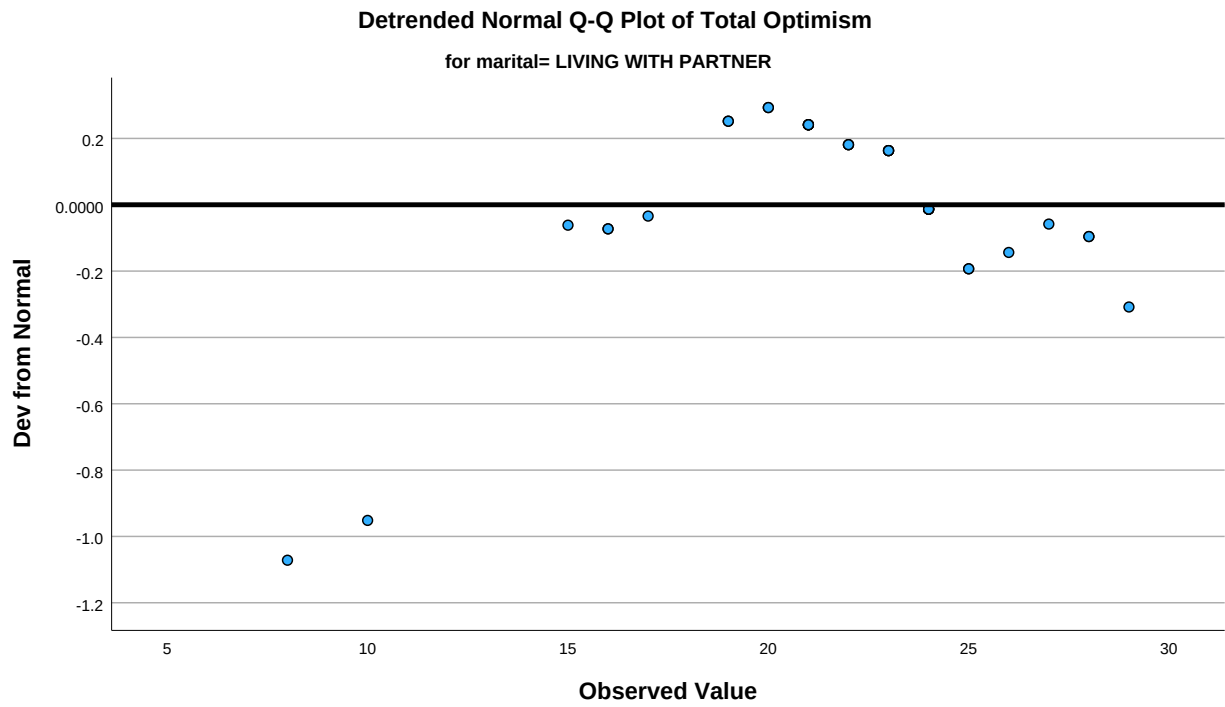


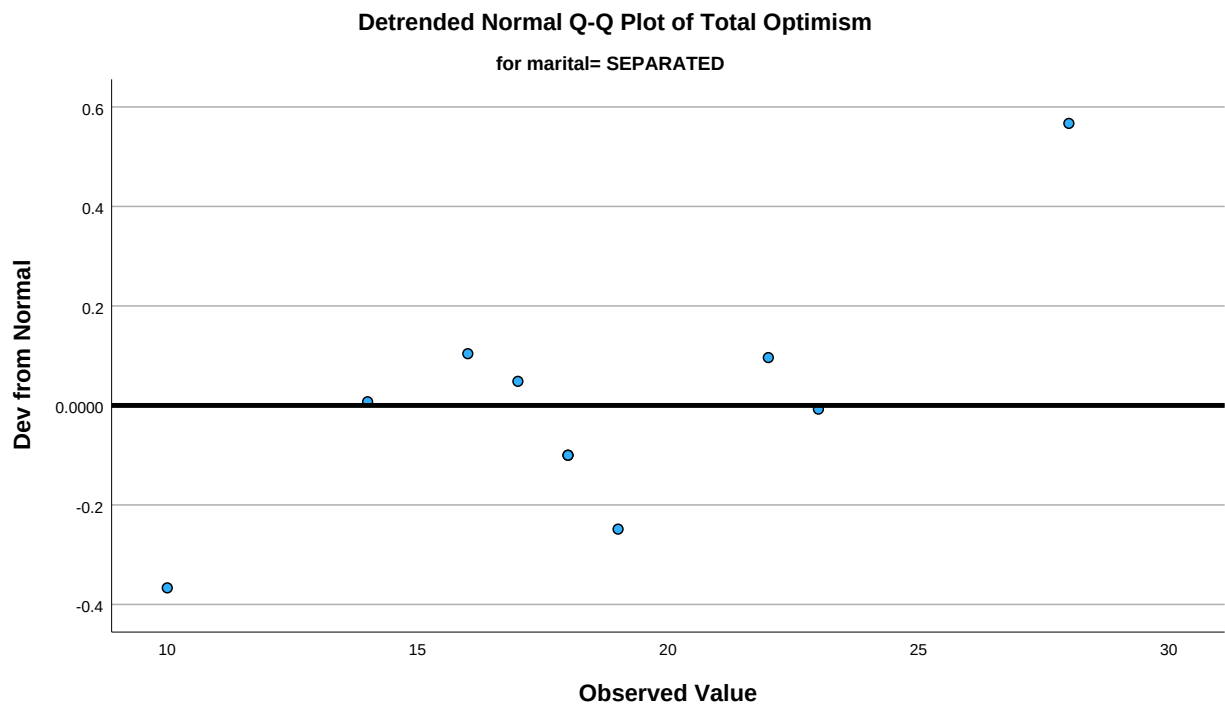
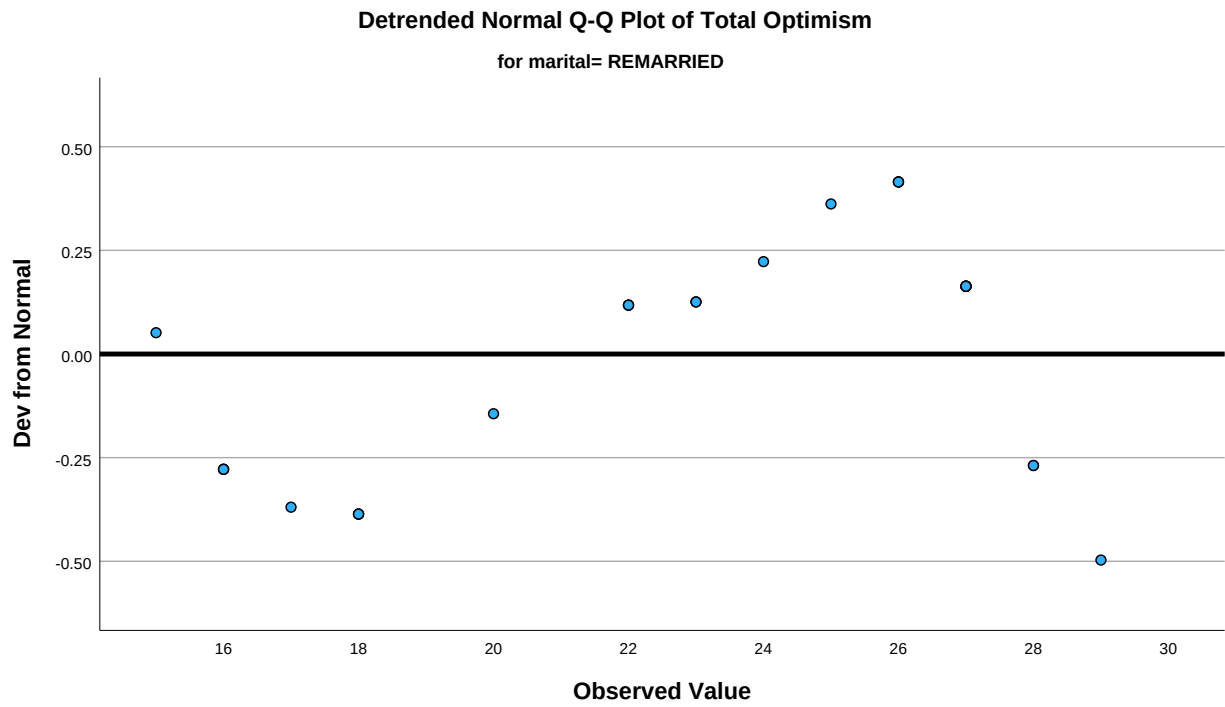


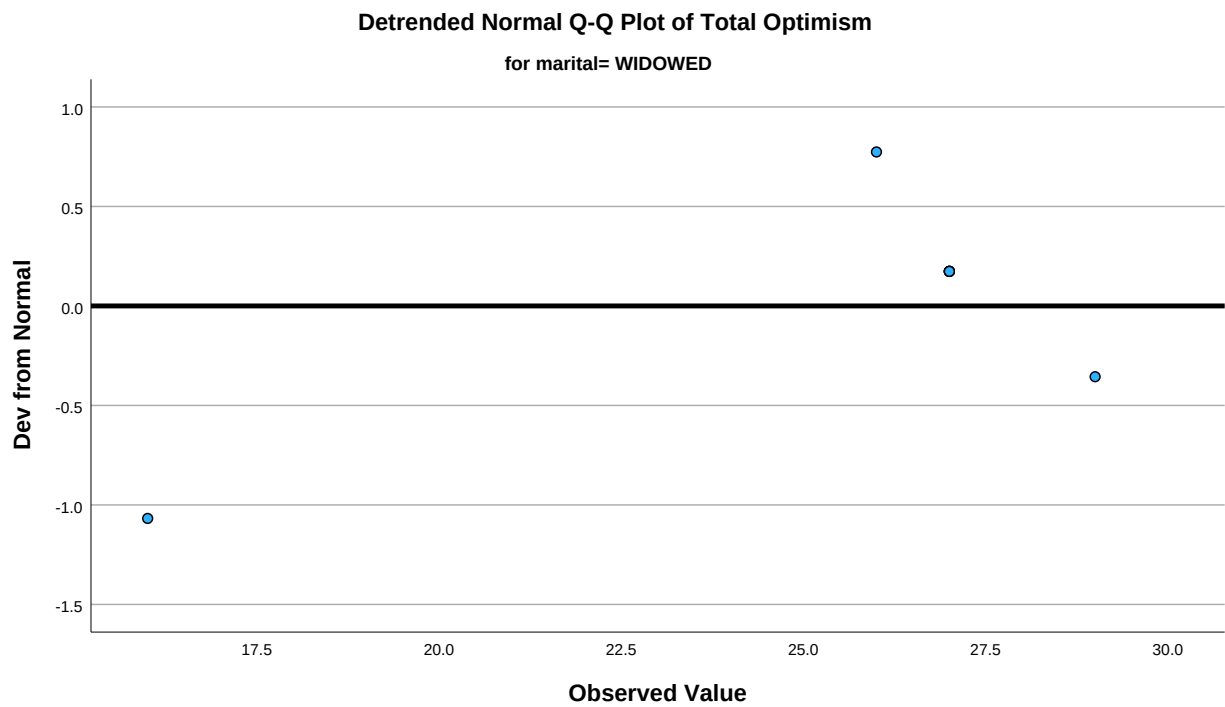
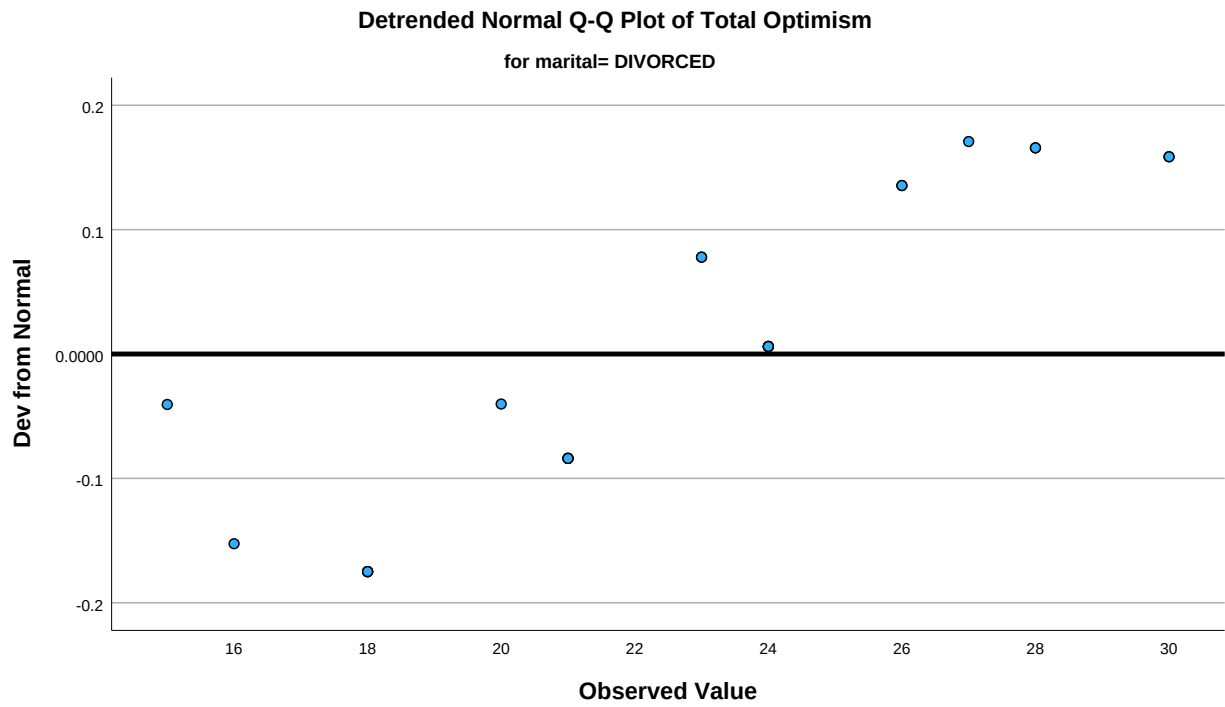


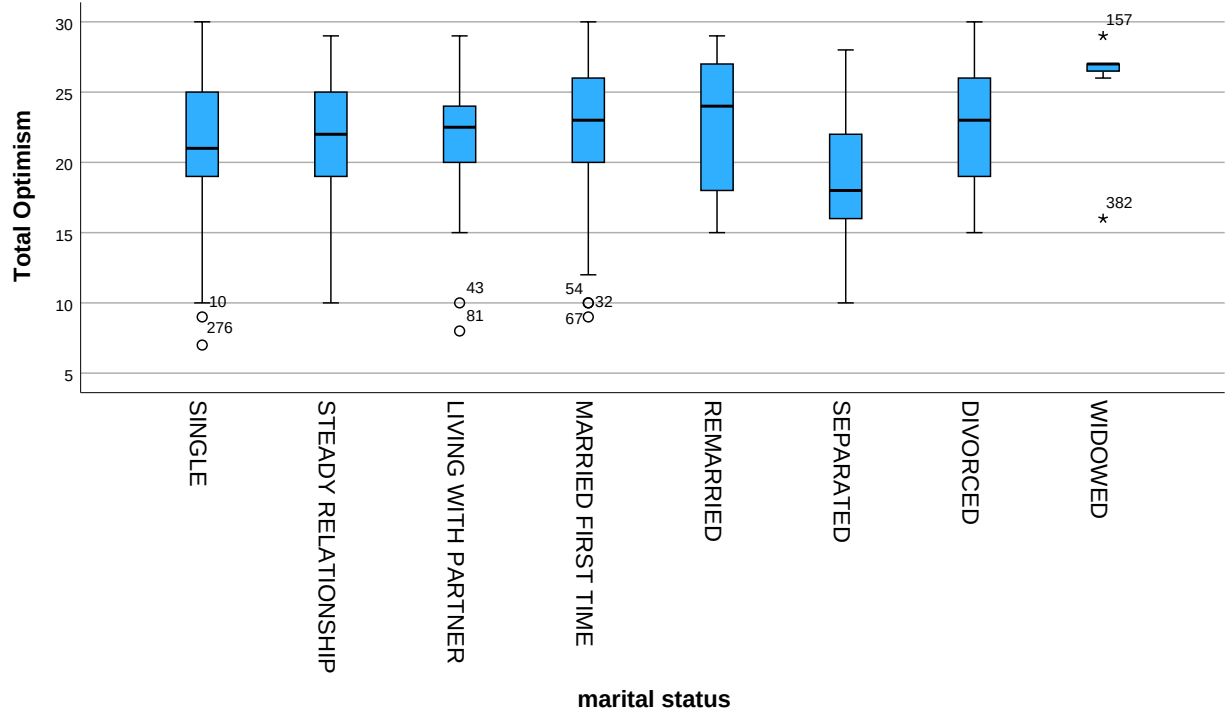
Detrended Normal Q-Q Plots











Normality check for Optimism across age group and optimism

Notes

Output Created		06-NOV-2024 18:04:24
Comments		
Input	Data	D:\survey.sav
	Active Dataset	DataSet2
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	439
Missing Value Handling	Definition of Missing	User-defined missing values for dependent variables are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.
Syntax		EXAMINE VARIABLES=toptim BY smoke agegp3 /PLOT BOXPLOT STEMLEAF HISTOGRAM NPLOT /COMPARE GROUPS /STATISTICS DESCRIPTIVES /INTERVAL 95 /MISSING LISTWISE /NOTOTAL.
Resources	Processor Time	00:00:03.06
	Elapsed Time	00:00:02.46

smoker

Case Processing Summary

		Valid		Cases Missing		Total	
smoker		N	Percent	N	Percent	N	Percent
Total Optimism	YES	85	100.0%	0	0.0%	85	100.0%
	NO	347	98.9%	4	1.1%	351	100.0%

Descriptives

smoker		Statistic		Std. Error
Total Optimism	YES	Mean	21.51	.520
		95% Confidence Interval for Mean	Lower Bound	20.47
			Upper Bound	22.54
		5% Trimmed Mean	21.65	
		Median	23.00	
		Variance	22.991	
		Std. Deviation	4.795	
		Minimum	9	
		Maximum	30	
		Range	21	
		Interquartile Range	7	
		Skewness	-.442	.261
		Kurtosis	-.230	.517
	NO	Mean	22.29	.233
		95% Confidence Interval for Mean	Lower Bound	21.83
			Upper Bound	22.75
		5% Trimmed Mean	22.45	
		Median	22.00	
		Variance	18.818	
		Std. Deviation	4.338	
		Minimum	7	
		Maximum	30	
		Range	23	
		Interquartile Range	6	
		Skewness	-.506	.131
		Kurtosis	.358	.261

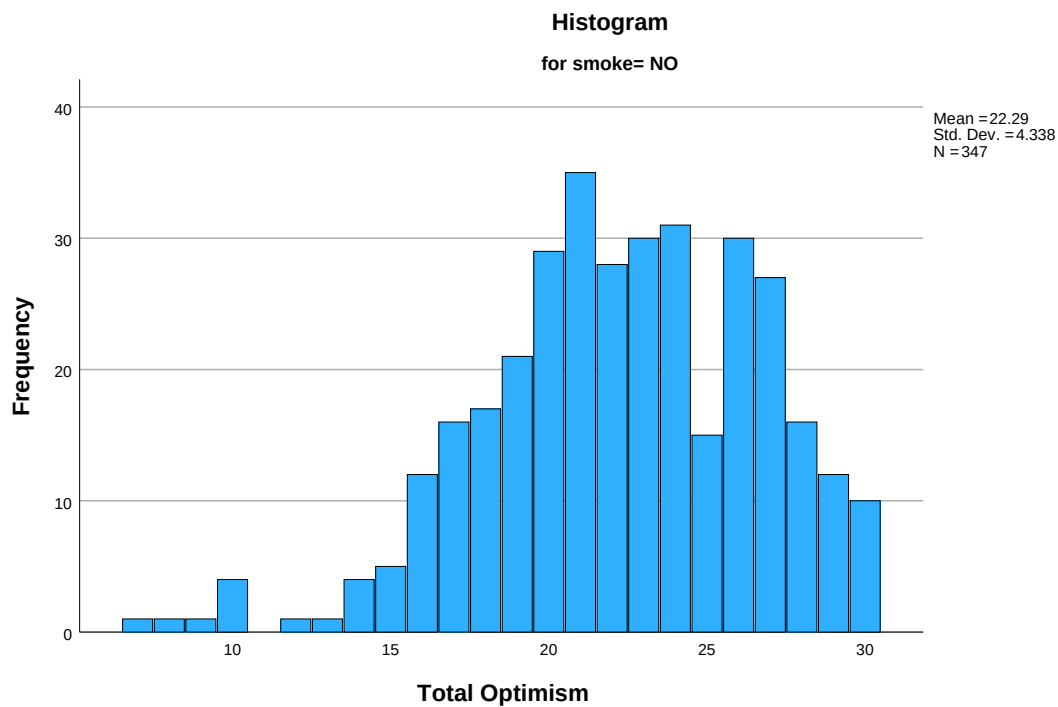
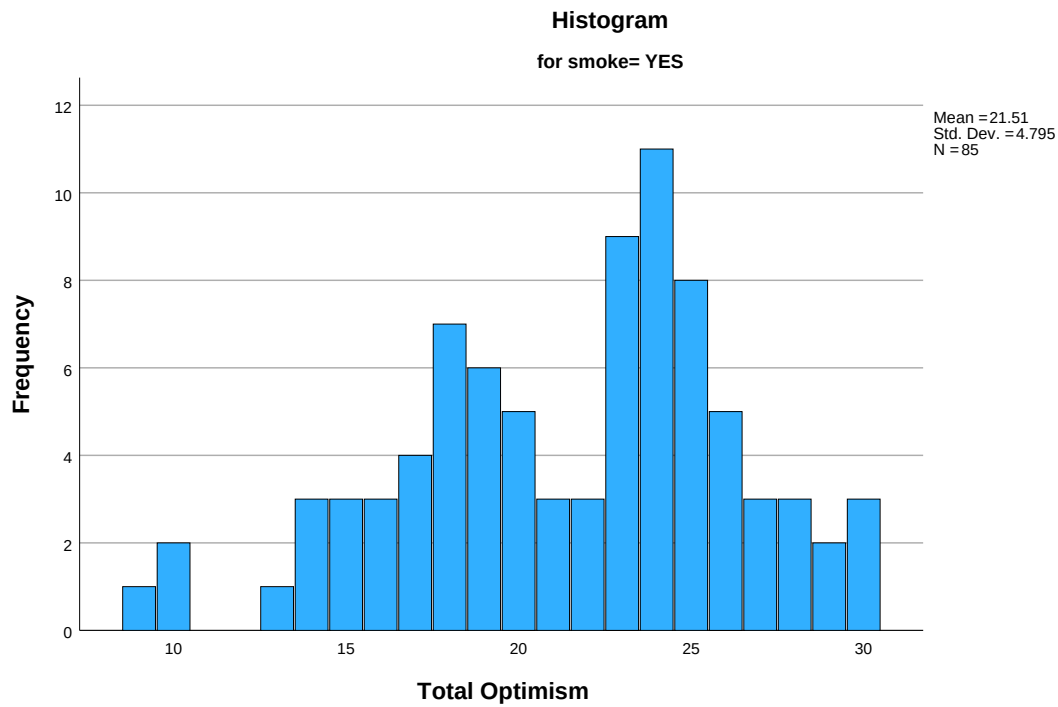
Tests of Normality

		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Total Optimism	YES	.140	85	<.001	.970	85	.044
	NO	.078	347	<.001	.973	347	<.001

a. Lilliefors Significance Correction

Total Optimism

Histograms



Stem-and-Leaf Plots

Total Optimism Stem-and-Leaf Plot for
smoke= YES

Frequency Stem & Leaf

```

1.00      0 . 9
6.00      1 . 003444
23.00     1 . 5556667777888888999999
31.00     2 . 0000011122233333334444444444
21.00     2 . 55555556666677788899
3.00      3 . 000

```

Stem width: 10
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
smoke= NO

Frequency Stem & Leaf

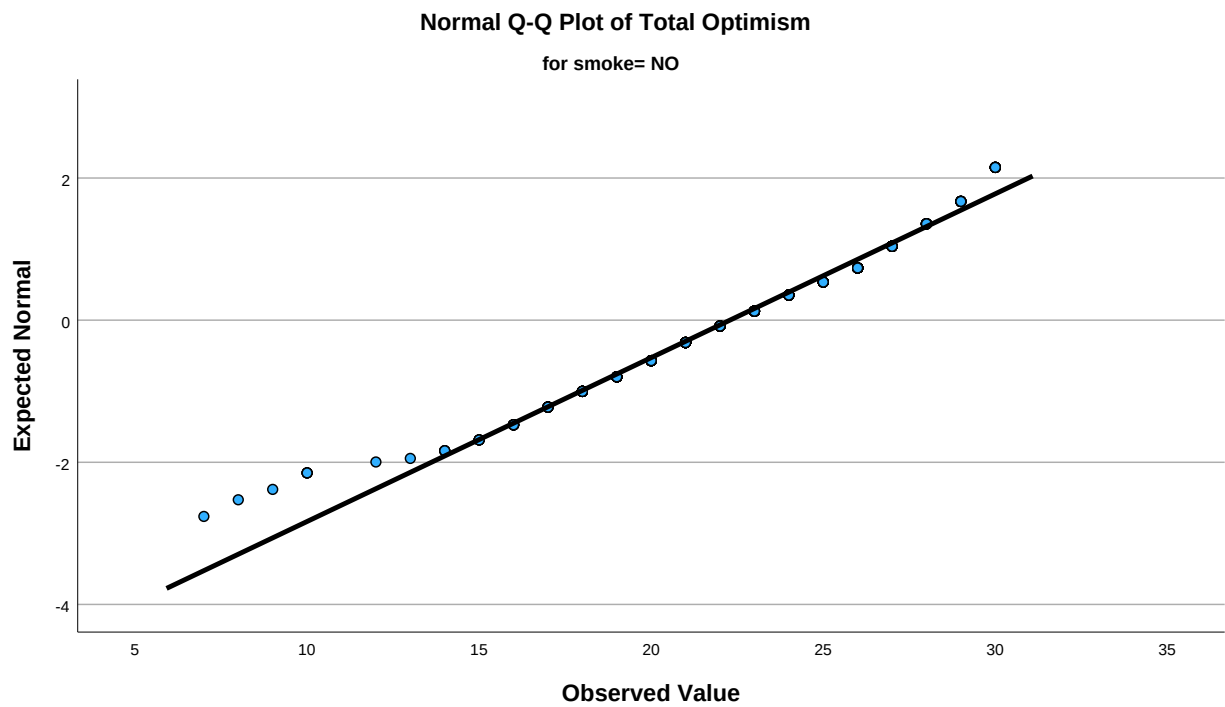
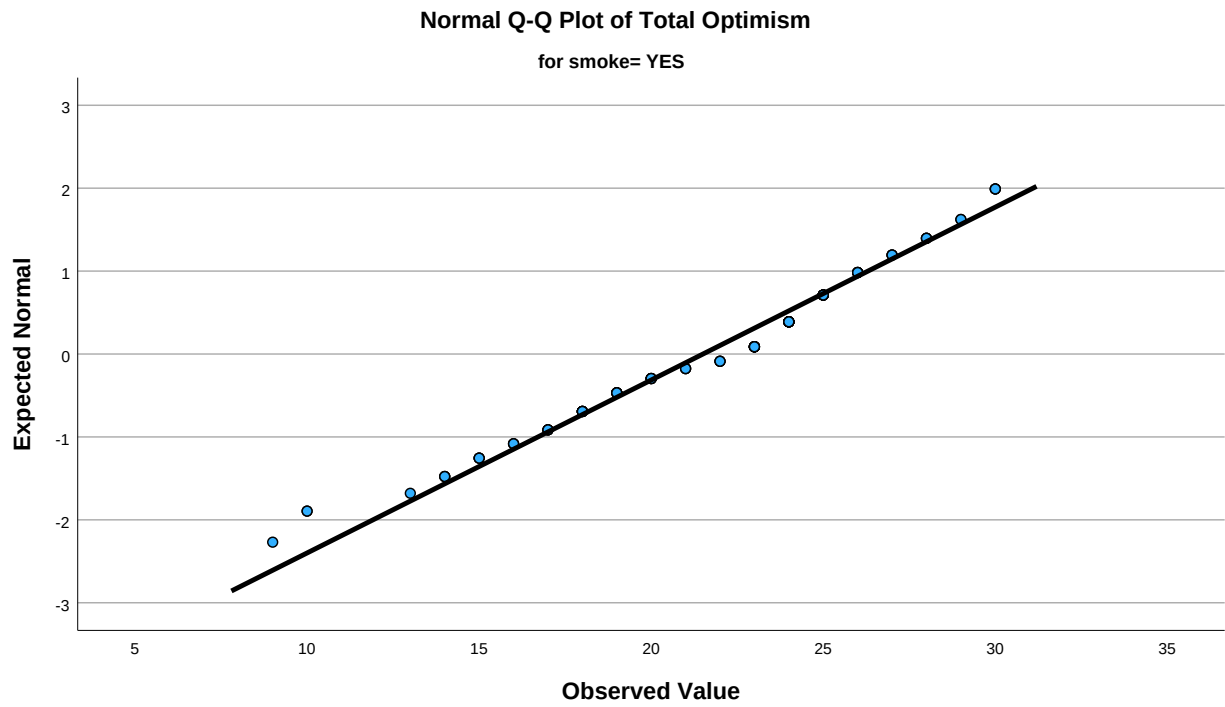
```

7.00 Extremes    (<=10.0)
1.00      12 . 0
1.00      13 . 0
4.00      14 . 0000
5.00      15 . 00000
12.00     16 . 000000000000
16.00     17 . 0000000000000000
17.00     18 . 0000000000000000
21.00     19 . 00000000000000000000
29.00     20 . 0000000000000000000000000000
35.00     21 . 0000000000000000000000000000000000
28.00     22 . 0000000000000000000000000000
30.00     23 . 000000000000000000000000000000
31.00     24 . 000000000000000000000000000000
15.00     25 . 0000000000000000
30.00     26 . 000000000000000000000000000000
27.00     27 . 0000000000000000000000000000
16.00     28 . 0000000000000000
12.00     29 . 000000000000
10.00     30 . 0000000000

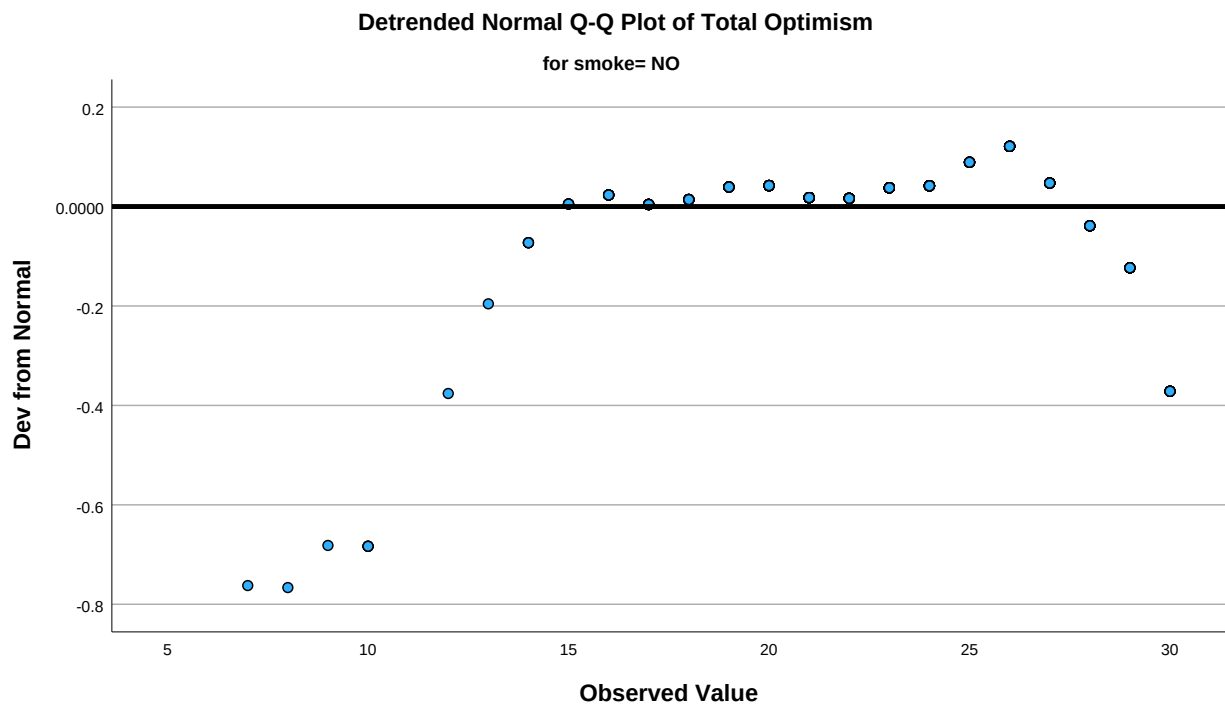
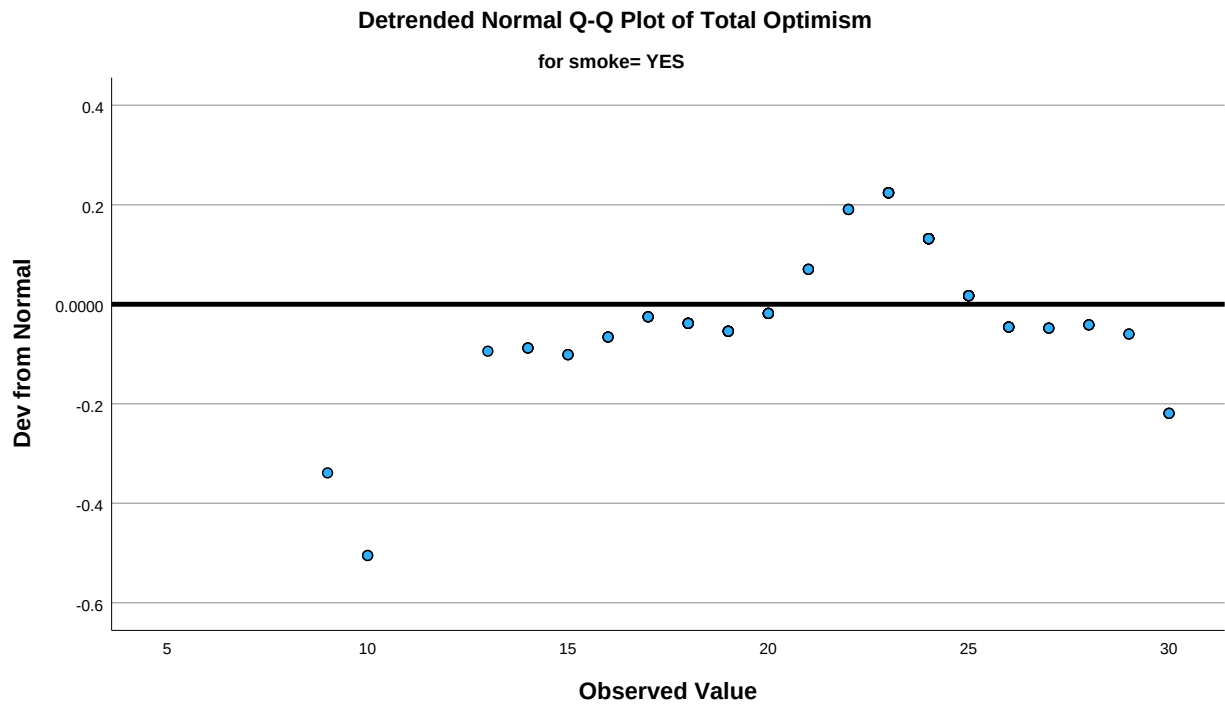
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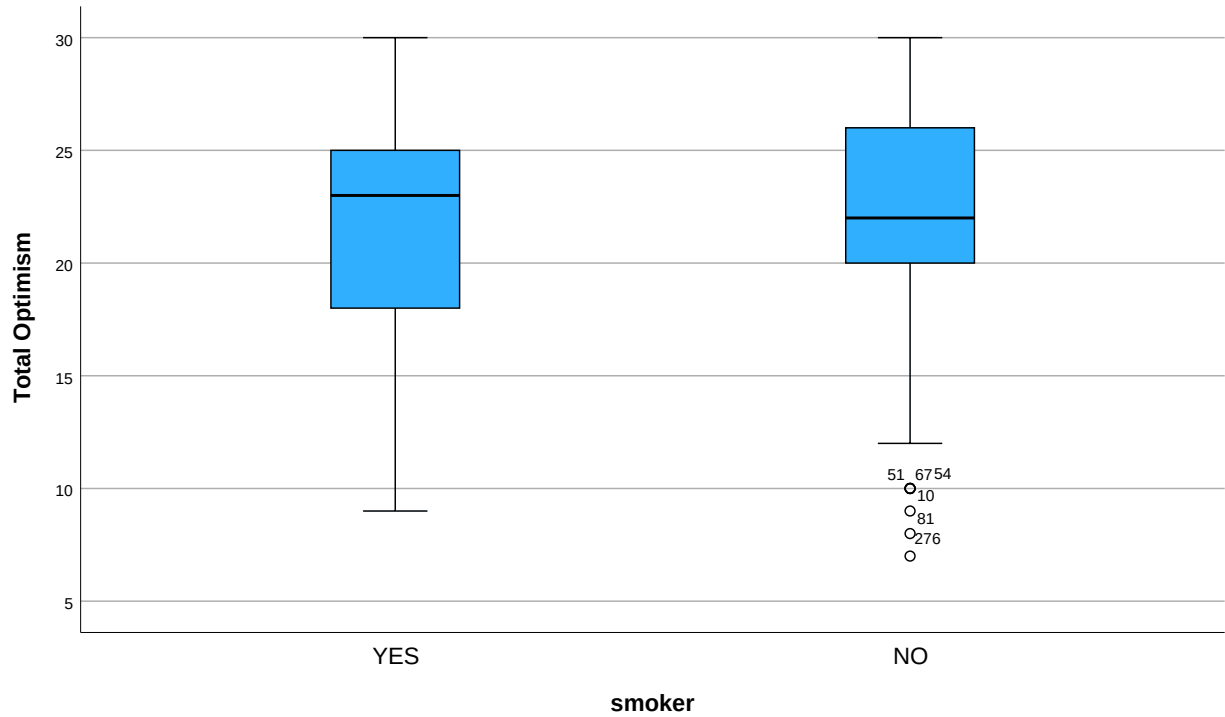
Stem width: 1
Each leaf: 1 case(s)

Normal Q-Q Plots



Detrended Normal Q-Q Plots





age 3 groups

Case Processing Summary

		Valid		Cases Missing		Total	
age 3 groups		N	Percent	N	Percent	N	Percent
Total Optimism	18 - 29	146	98.0%	3	2.0%	149	100.0%
	30 - 44	153	100.0%	0	0.0%	153	100.0%
	45+	133	97.1%	4	2.9%	137	100.0%

Descriptives

age 3 groups		Statistic		Std. Error
Total Optimism	18 - 29	Mean	21.36	.378
		95% Confidence Interval for Mean	Lower Bound	20.62
			Upper Bound	22.11
		5% Trimmed Mean	21.56	
		Median	22.00	
		Variance	20.854	
		Std. Deviation	4.567	
		Minimum	7	
		Maximum	30	
		Range	23	
		Interquartile Range	5	
		Skewness	-.610	.201
		Kurtosis	.572	.399
	30 - 44	Mean	22.10	.335
		95% Confidence Interval for Mean	Lower Bound	21.44
			Upper Bound	22.77
		5% Trimmed Mean	22.23	
		Median	22.00	
		Variance	17.200	
		Std. Deviation	4.147	
		Minimum	10	
		Maximum	30	
		Range	20	
		Interquartile Range	6	
		Skewness	-.400	.196
		Kurtosis	.059	.390
	45+	Mean	23.02	.389
		95% Confidence Interval for Mean	Lower Bound	22.25
			Upper Bound	23.78
		5% Trimmed Mean	23.17	
		Median	24.00	
		Variance	20.121	
		Std. Deviation	4.486	
		Minimum	8	
		Maximum	30	
		Range	22	
		Interquartile Range	7	
		Skewness	-.518	.210
		Kurtosis	-.120	.417

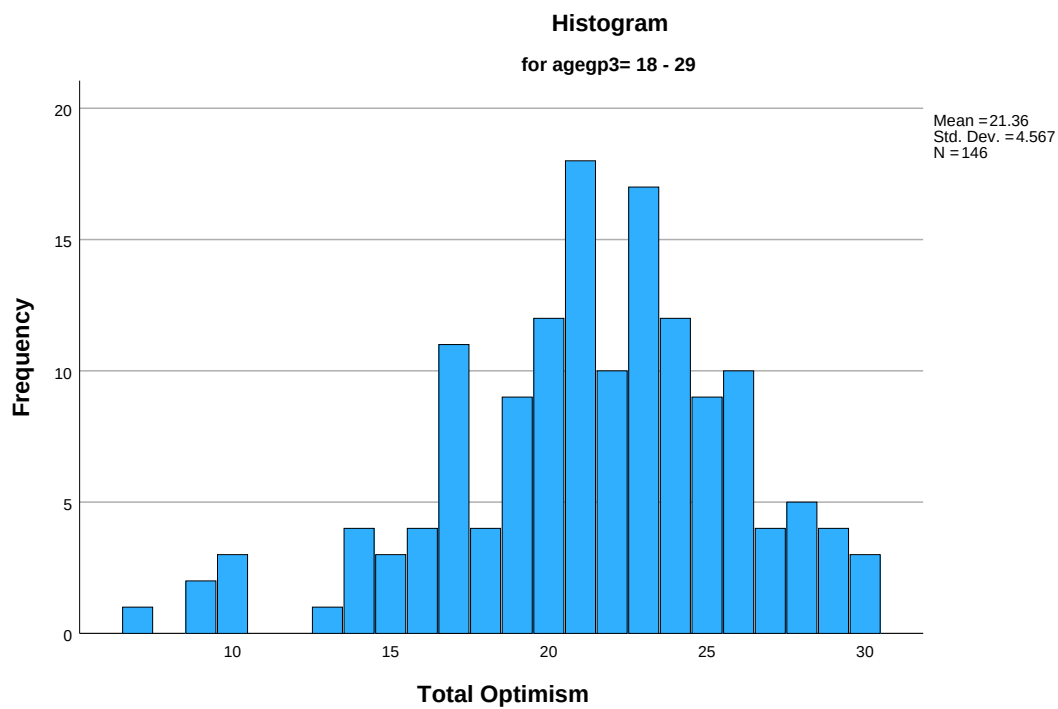
Tests of Normality

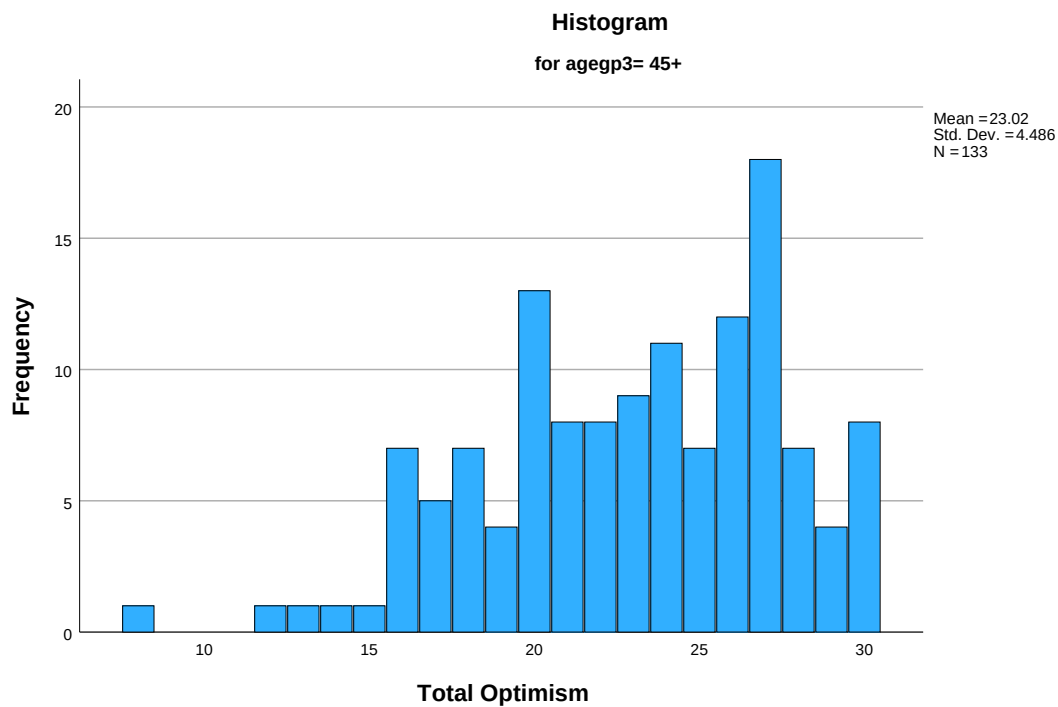
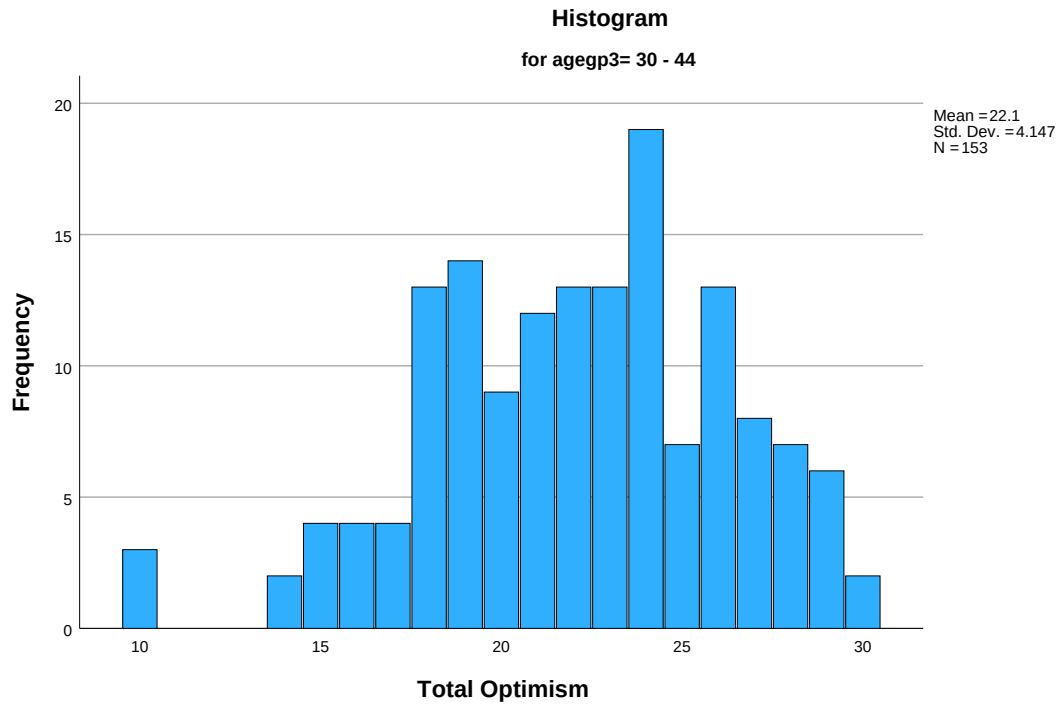
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	age 3 groups	Statistic	df	Sig.	Statistic	df	Sig.
Total Optimism	18 - 29	.098	146	.001	.969	146	.002
	30 - 44	.081	153	.015	.976	153	.009
	45+	.116	133	<.001	.962	133	<.001

a. Lilliefors Significance Correction

Total Optimism

Histograms





Stem-and-Leaf Plots

Total Optimism Stem-and-Leaf Plot for
agegp3= 18 - 29

Frequency Stem & Leaf

6.00 Extremes (≤ 10.0)

1.00	13 . 0
4.00	14 . 0000
3.00	15 . 000
4.00	16 . 0000
11.00	17 . 0000000000
4.00	18 . 0000
9.00	19 . 00000000
12.00	20 . 0000000000
18.00	21 . 0000000000000000
10.00	22 . 0000000000
17.00	23 . 0000000000000000
12.00	24 . 0000000000
9.00	25 . 00000000
10.00	26 . 0000000000
4.00	27 . 0000
5.00	28 . 00000
4.00	29 . 0000
3.00	30 . 000

Stem width: 1
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
agegp3= 30 - 44

Frequency Stem & Leaf

3.00	1 . 000
.00	1 .
6.00	1 . 44555
8.00	1 . 6667777
27.00	1 . 88888888888889999999999999
21.00	2 . 00000000111111111111
26.00	2 . 222222222222333333333333
26.00	2 . 444444444444444444445555555
21.00	2 . 66666666666677777777
13.00	2 . 888888999999
2.00	3 . 00

Stem width: 10
Each leaf: 1 case(s)

Total Optimism Stem-and-Leaf Plot for
agegp3= 45+

Frequency Stem & Leaf

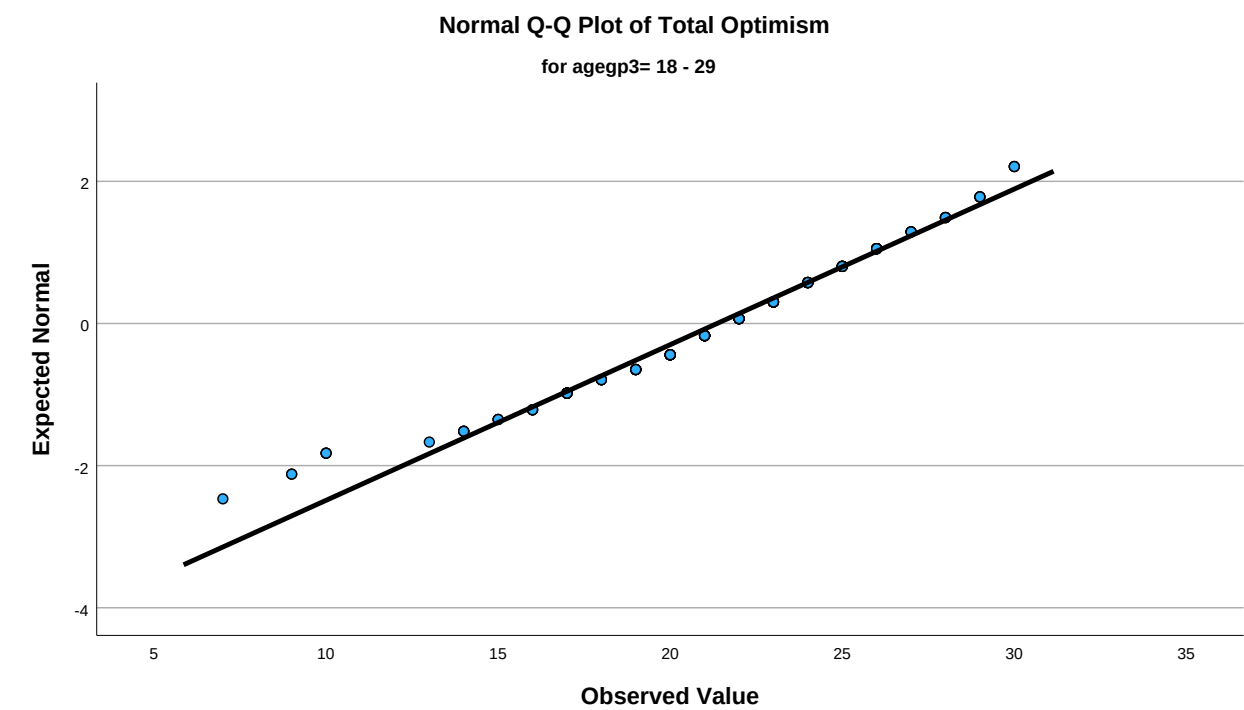
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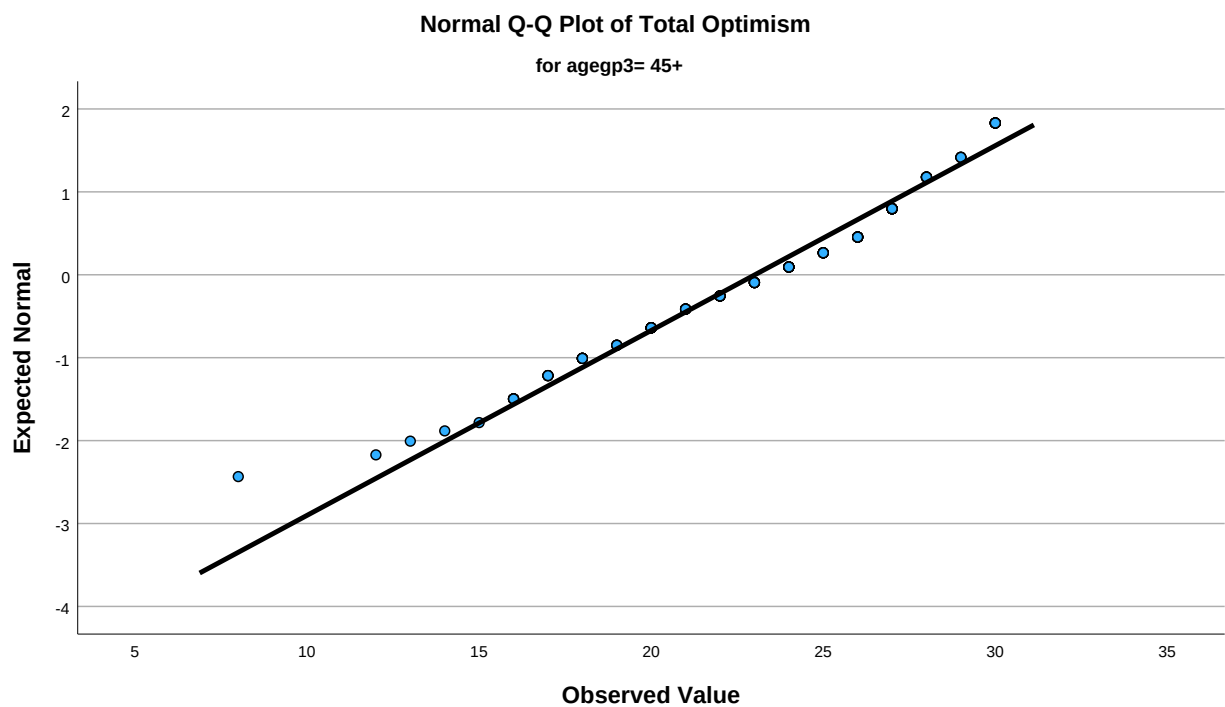
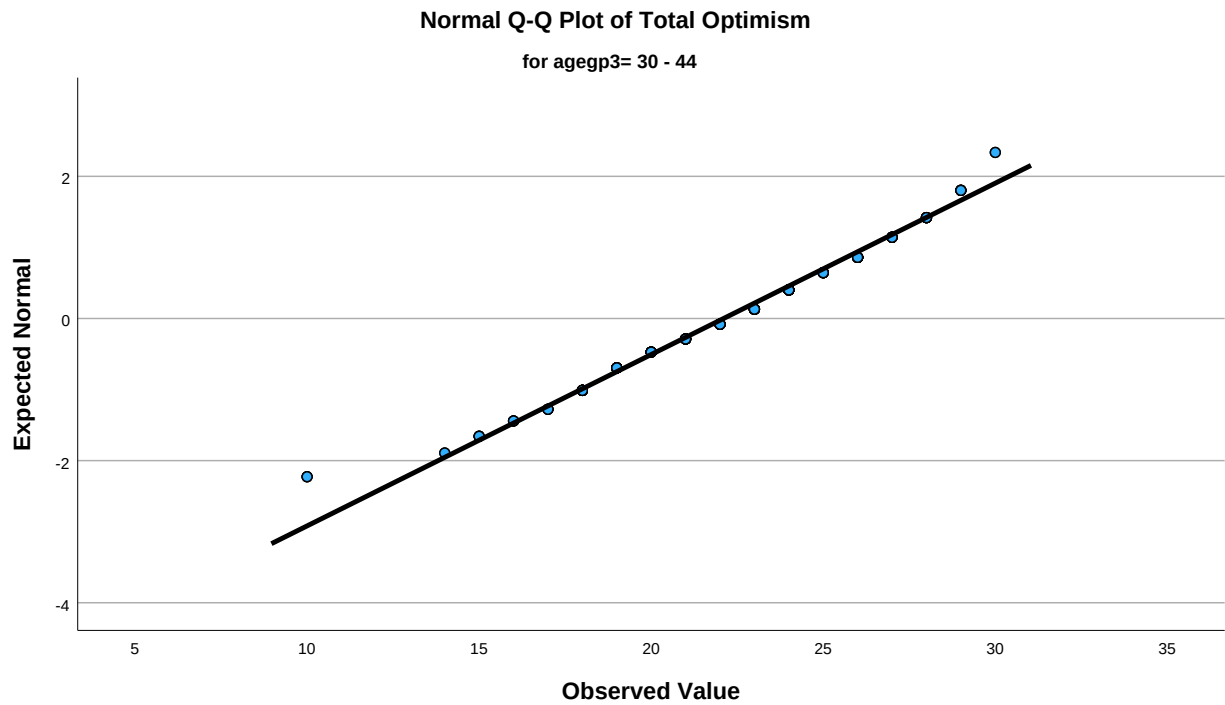
1.00 Extremes    (=8.0)
1.00  12 . 0
1.00  13 . 0
1.00  14 . 0
1.00  15 . 0
7.00  16 . 0000000
5.00  17 . 00000
7.00  18 . 0000000
4.00  19 . 0000
13.00  20 . 000000000000000
8.00  21 . 00000000
8.00  22 . 00000000
9.00  23 . 000000000
11.00  24 . 00000000000
7.00  25 . 0000000
12.00  26 . 0000000000000
18.00  27 . 0000000000000000000
7.00  28 . 0000000
4.00  29 . 0000
8.00  30 . 00000000

Stem width:      1
Each leaf:      1 case(s)

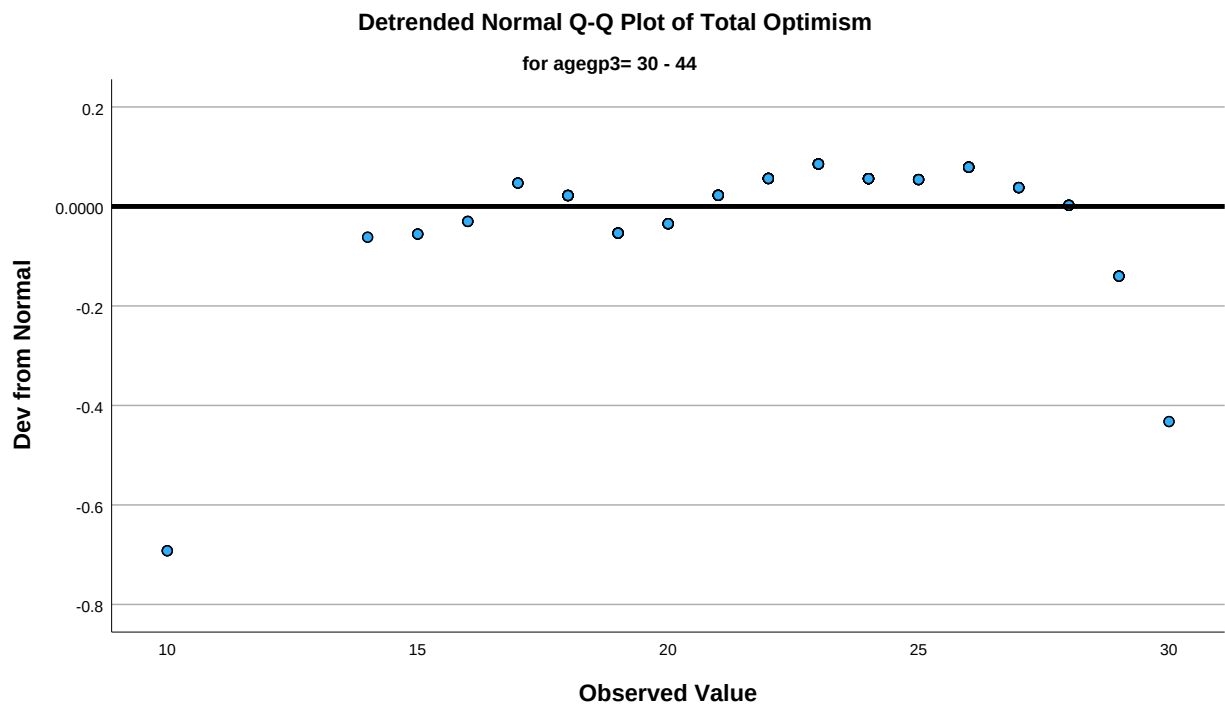
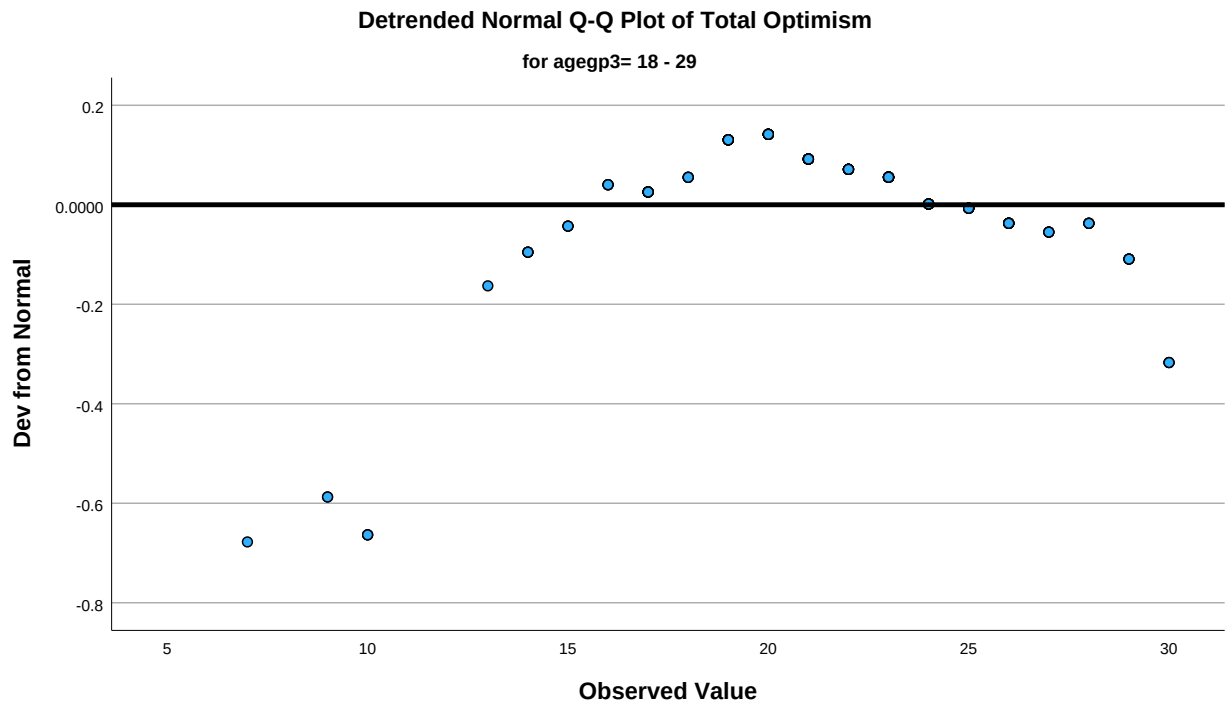
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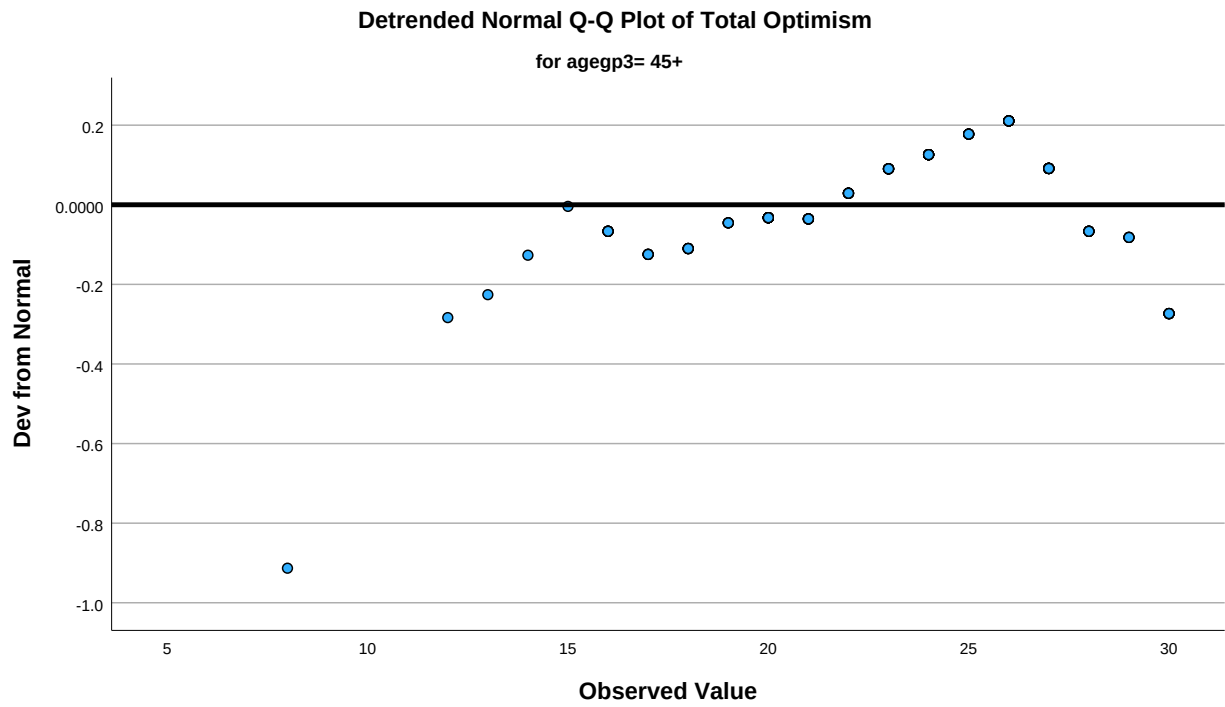
Normal Q-Q Plots





Detrended Normal Q-Q Plots





Boxplots

